

3.2 ATTAINMENT OF COURSE OUTCOMES (75)

3.2.1 Describe the assessment tools and processes used to gather the data upon which the evaluation of Course Outcome is based (10)

The document outlines the process for assessing Course Outcomes (COs) in an academic course, detailing the weightage given to different assessment methods. Here's a detailed explanation of the process:

1. Course Outcomes (COs) Assessment:

Direct Attainment (80%): This refers to the assessment methods that directly measure students' achievement of the course outcomes. It includes internal assessments and end-semester examinations.

Indirect Attainment (20%): This involves methods that indirectly measure students' perception of their learning and the course's effectiveness, typically through course end surveys.

2. Direct Assessment (80%):

Internal Assessments (20%): These are conducted throughout the course and include multiple internal examinations (IA1, IA2, IA3) and assignments. The internal assessments contribute 20% to the overall course assessment.

End Semester Examinations (80%): These are comprehensive exams conducted at the end of the semester, contributing 80% to the overall course assessment.

3. Conversion of Marks:

The total marks from internal assessments and end-semester examinations are converted to fit the 20% and 80% weightage, respectively, within the direct assessment component.

4. Indirect Assessment (20%):

This is conducted through course end surveys where students provide feedback on their learning experience and the course's effectiveness. This feedback is used to gauge the indirect attainment of course outcomes.

5. Practical Examination or Project Evaluation:

For courses with practical components, the assessment includes lab conduction and assessments or project reviews (20%) and end-semester practical examinations (80%).

In summary, the CO assessment is a balanced approach that combines direct and indirect methods to evaluate students' learning outcomes. Direct methods focus on academic performance through examinations and assignments, while indirect methods gather student feedback to assess the course's impact on their learning. This comprehensive approach ensures a thorough evaluation of both student performance and course effectiveness.

CO Assessment Tool

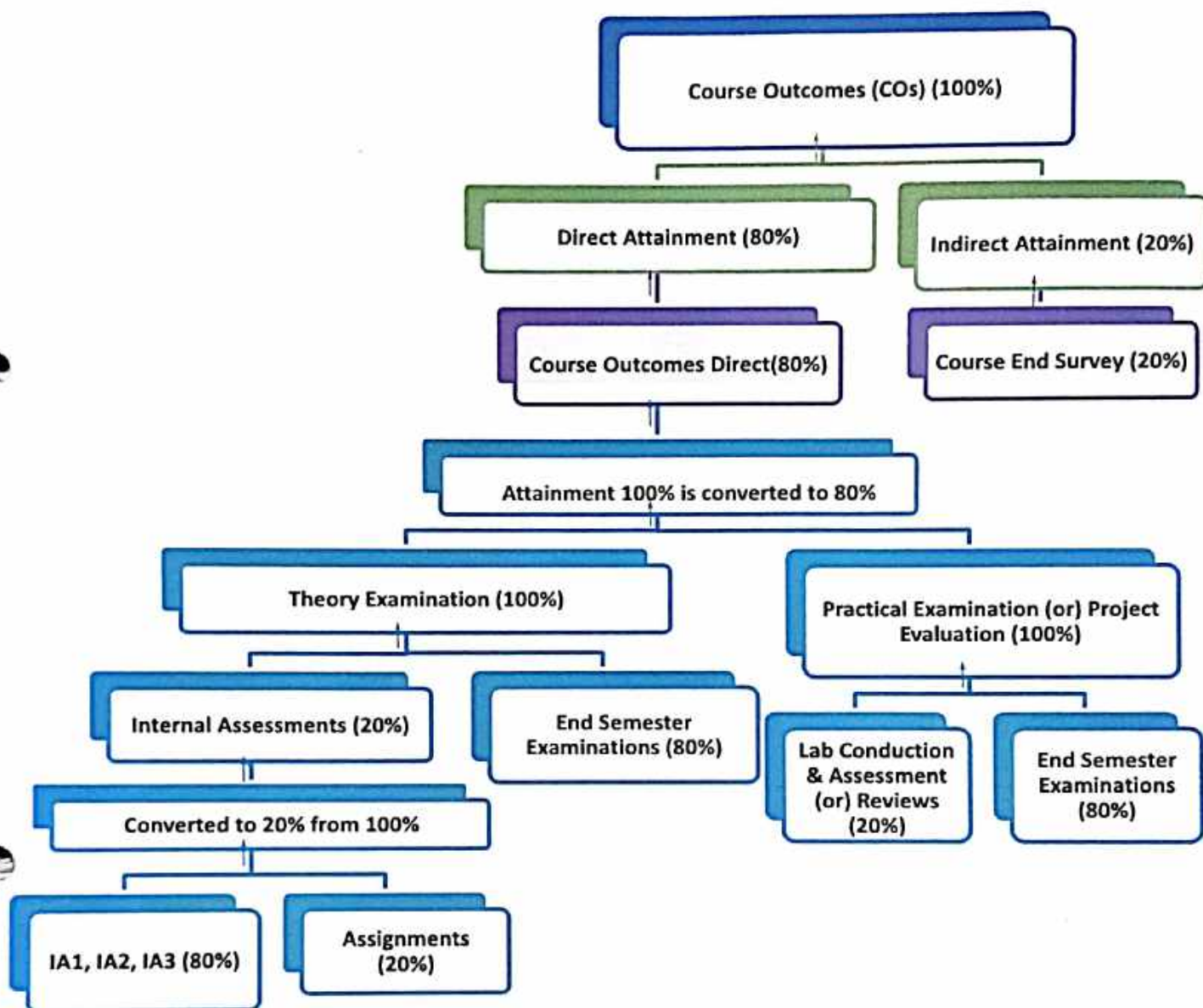
The CO Assessment Tool is a structured mechanism designed to evaluate the achievement of Course Outcomes (COs) in an academic program. It encompasses a variety of methods and instruments, such as quizzes, exams, assignments, projects, presentations, and practical demonstrations, tailored to measure specific learning objectives defined for a course. The tool ensures that assessments are aligned with the COs, providing a clear and measurable way to gauge student performance. It often includes rubrics, scoring guides, or performance indicators to maintain consistency and fairness in evaluation. By using this tool, educators can systematically collect data on student learning, analyze attainment levels, and identify areas for improvement in course delivery and curriculum design.

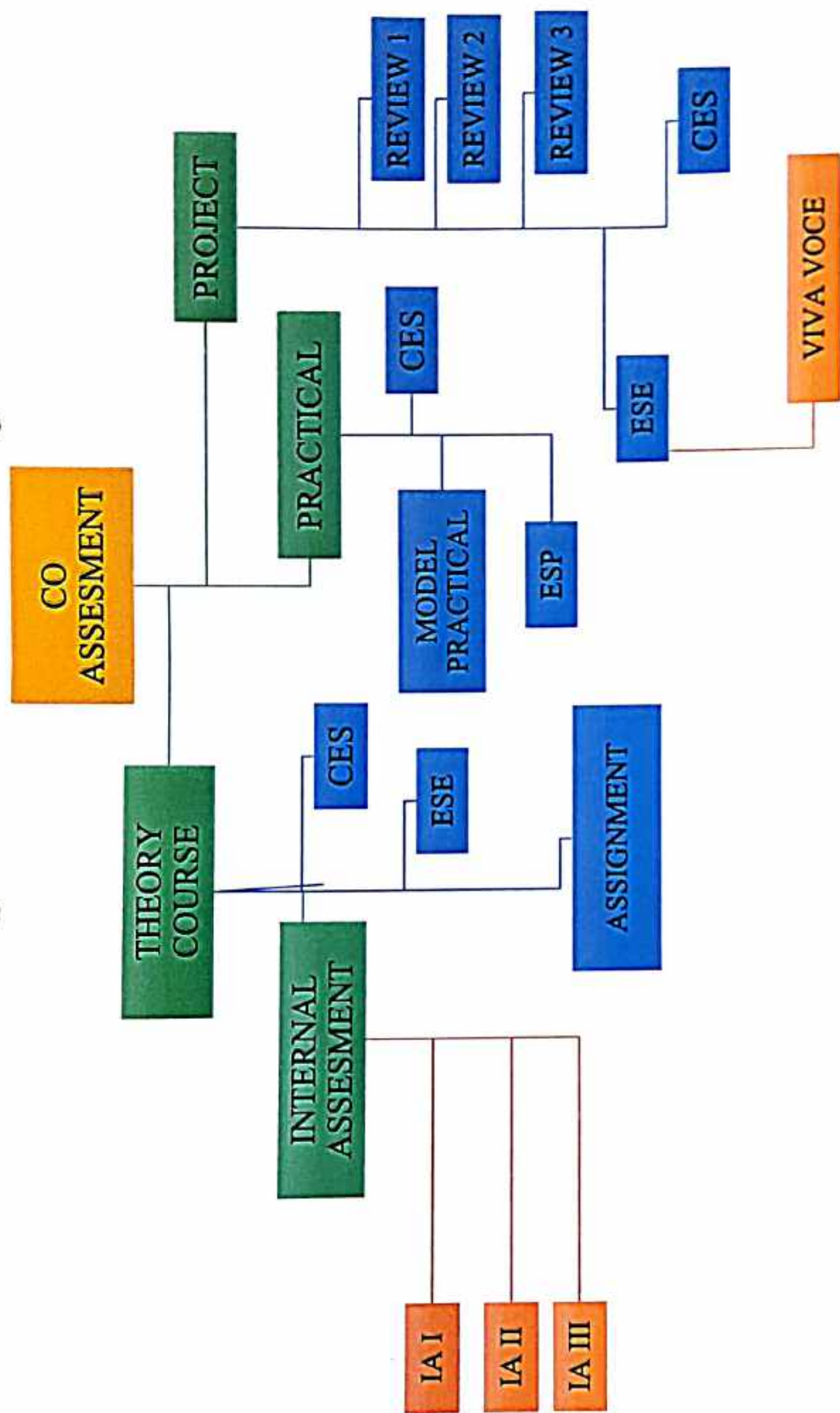
Tools used in measuring Course Outcomes

The method of assessment method includes:

- Internal Assessments (IA-I, IA-II and IA-III)
- End Semester Examination
- Practical Examination
- Project Evaluation
- Assignments
- Course End Survey

Assessment Tools for the Evaluation of Course Outcomes





CO ASSESSMENT TOOLS

S.A Engineering College, Chennai -77
Department of Computer Science & Engineering
Assessment tools

Type of Assessment	Assessment Tool / Method	Purpose & Process Description	Frequency	Weightage / Contribution
DIRECT ASSESSMENT (CO Level)	Internal Assessments (IA I, II, III)	Evaluate understanding of course content through periodic tests (Unit-wise & comprehensive). Used for teaching improvement and CO mapping.	Thrice per semester	20% (of CO Direct)
	End Semester Examination	Summative evaluation covering entire syllabus; measures final learning achievement.	Once per semester	80% (of CO Direct)
	Assignments	Reinforce classroom learning and assess self-learning, analytical, and problem-solving skills.	Minimum 3 per course	Included in 20% internal
	Practical Examination / Lab Evaluation	Assess practical skills, preparation, observation, analysis, and report writing. Includes model exam and final practical.	Once per semester	Internal 20%, External 80%
	Project Evaluation (Final Year)	Evaluate project work through 3 internal reviews and external viva; rubrics-based assessment on innovation, implementation, teamwork, and presentation.	Thrice + final viva	Internal 20%, External 80%
INDIRECT ASSESSMENT (CO Level)	Course End Survey	Student feedback on teaching quality, clarity, examples, and alignment of assessments with COs. Used for continuous improvement.	End of each semester	20% of CO (Indirect)
PROGRAM LEVEL (PO / PSO ATTAINMENT)	Direct Assessment (via COs)	PO/PSO attainment derived from CO attainments and CO-PO mapping using correlation matrix.	End of each semester	70% of total PO/PSO
	Student Exit Survey	Feedback from graduating students on overall program	End of program	10% of Indirect PO/PSO

		quality, skill acquisition, and career readiness.		
	Alumni Survey	Collects input from alumni on curriculum relevance and industry readiness.	Periodically	7% of Indirect PO/PSO
	Employer Survey	Gathers feedback from employers regarding graduates' performance, technical skills, and competencies.	Annually	3% of Indirect PO/PSO
	ECA / CCA Event Feedback	Evaluates impact of co-curricular and extracurricular activities (seminars, workshops, internships, etc.) on PO/PSO attainment.	Periodically	10% of Indirect PO/PSO
TOTAL ATTAINMENT FORMULA	For COs: Total = $(IA \times 0.2) + (\text{End Sem} \times 0.8)$ For POs/PSOs: = $(\text{Direct} \times 0.7) + (\text{Indirect} \times 0.3)$	Calculation combines all assessment data to derive final attainment levels.	End of academic year	—

S.A ENGINEERING COLLEGE, CHENNAI-77**Department of Computer Science & Engineering****Rubrics for Internal Assessments****Register Number:****Degree/Branch/Class:****Date & Session:****Subject Code/Subject Title:****Academic Year/Semester:****Internal Assessment: IA1/IA2/IA3****Evaluated by:**

Trait	Unsatisfactory	Satisfactory	Good	Excellent
Terminology	Terminologies not defined or defined irrelevant to the question but not explained or explained incorrectly.	Few Terminologies are correctly defined with incorrect explanations or not explained.	All Terminologies defined relevant to the question with limited explanations.	All Terminologies defined, relevant to the question and well explained.
Diagram	Irrelevant, Not accurate and wrongly labeled with incorrect explanations or not explained.	Relevant but not accurate, explanations are incorrect or not explained.	Relevant diagrams are included with limited explanations.	Relevant, accurate and correctly labeled diagrams with full explanations.
Explanation	Shows no understanding of the topic and no argument developed relevant to the question.	Shows a superficial understanding of the topic, argument not developed enough to the question.	Shows a limited understanding of the topic, not quite a fully developed argument.	Shows a deep/robust understanding of the topic with a fully developed argument.
Problem Solving	Unable to solve the problem with incorrect explanation or no explanation.	Unable to solve all parts of the problem with less explanation.	Able to solve using only one strategy with explanation.	Able to solve with full understanding of the problem using multiple strategies with detailed and thorough explanation.
Neat Presentation	The work appears sloppy and unorganized. It is hard to know what information goes together.	- The work is presented in a organized fashion that is not easy to read	The work is presented in a neat, organized fashion that is easy to read	The work is presented in a neat, clear, organized fashion that is easy to read

Excellent: (81-100 Marks) S Grade/A Grade**Good: (61-80 Marks) B Grade/ C Grade****Satisfactory: (50-60), E Grade (50-55)****Unsatisfactory: (<50) U Grade****Grade secured: S / A / B / C / D / E / U**

S.A ENGINEERING COLLEGE, CHENNAI-77

Department of Computer Science & Engineering

Rubrics for Assignment

Register Number:

Subject Code/Subject Title:

Degree/Branch/Class:

Academic Year/Semester:

Date of submission:

Assignment No.: A1/A2/A3

Evaluated by:

Trait	Unsatisfactory (00-03 Marks)	Satisfactory (04-06 Marks)	Good (07-09 Marks)	Excellent (10 Marks)
Relation of Assignment to Course Outcomes	Assignment is not related to Course Outcomes.	Assignment is somehow related to Course Outcomes.	Assignment is related to Course Outcomes.	Assignment is clearly related to Course Outcomes.
Challenging and Rewarding	Understanding is very poor.	Understanding is reasonably good.	Challenging and Rewarding.	Challenging and Rewarding.
Analysis and Originality	Simplistic view of the topic with no alternative views. Writing is more descriptive than analytical.	Even balance between critical thinking and description. Even balance between critical thinking and description.	Effective use of critical thinking in analysis. Effective use of critical thinking in analysis.	Analysis is fresh, posing new ways to think of the material. Analysis is fresh, posing new ways to think of the material.
Submission of Assignment	Not submitted within the stipulated time.	Submitted few days after stipulated time.	Submitted next day after the stipulated time.	Submitted within the stipulated time.
Learning Outcomes	The learning outcomes for this assignment are poor.	The learning outcomes for this assignment are good.	The learning outcomes for this assignment are very good.	The learning outcomes for this assignment are excellent.

Grade secured: Excellent/Good/Satisfactory/Unsatisfactory

S.A ENGINEERING COLLEGE, CHENNAI-77

Department of Computer Science & Engineering

CO Attainment Rubrics

CO Attainment Level	Criteria	Attainment Range	Performance Description
Level 3 (High)	Target achieved by majority of students	$\geq 75\%$ of students	Excellent attainment, reflecting strong achievement of COs
Level 2 (Moderate)	Target partially achieved	$>65\% - \leq 75\%$ of students	Moderate attainment, reflecting acceptable achievement of COs
Level 1 (Low)	Target barely achieved	$>50\% - \leq 65\%$ of students	Low attainment, highlighting areas needing improvement
Level 0 (Very Low)	Target not achieved	$< 50\%$ of students	Very low attainment, indicating serious gaps in CO delivery

S.A ENGINEERING COLLEGE, CHENNAI-77

Department of Computer Science & Engineering

PO Attainment Rubrics

PO Attainment Level	Criteria	Attainment Range	Performance Description
Level 3 (High)	Weighted average of all COs mapped to the PO is high	$\geq 75\%$ of students	Excellent attainment, demonstrating strong achievement of the PO
Level 2 (Moderate)	Weighted average of all COs mapped to the PO is moderate	$>65\% - \leq 75\%$ of students	Moderate attainment, indicating acceptable achievement of the PO
Level 1 (Low)	Weighted average of all COs mapped to the PO is low	$>50\% - \leq 65\%$ of students	Low attainment, showing need for improvement in PO delivery
Level 0 (Very Low)	Weighted average of all COs mapped to the PO is very low	$< 50\%$ of students	Very low attainment, highlighting serious gaps in PO delivery

Department of Computer Science & Engineering

Rubrics for Assessing and Accepting Projects

Name & Register No. of Batchmates:

Degree/Branch/Class:

Date:

Evaluated by:

Subject Code/Subject Title:

Academic Year/Semester:

TRAIT	Unacceptable	Acceptable	Exemplary
Abstract	Vague about the purpose of investigation and/or does not provide a summary of the whole article.	States the purpose of the investigation in a vague way and summarizes the method, results, and conclusions sections.	In one or two paragraphs, clearly states the purpose of the investigation in the form of a problem, question, or issue to be resolved, and summarizes the method, results, and conclusions sections.
Introduction	Provides background research into the topic area but does not lay out a specific question to be answered or problem to be solved.	Provides background research into the topic area and puts a question to be answered or describes a problem to be solved in relation to this background.	Provides background research into the topic area summarizing important findings from earlier work with citations to books and articles and from recent periodicals and/or journals. Puts a question to be answered or describes a problem to be solved that clearly emerges from this background, and explains the significance of this problem for an audience of non-specialists.
Detailed analysis of Feasibility	Detailed and thorough explanation of specifications and Limitations.	Collects information and moderate study of the existing Systems.	Detailed and thorough explanation of specifications and Limitations.
Description of concepts and Technical Details	Inappropriate explanation of key concepts, poor description of technical requirements of the projects.	Inadequate description of technical requirements of the project.	Proper explanation of key concepts, strong description of technical requirements of the project.

Methods Results and Conclusions	Description of data analysis is either not present or poor. Description of research focuses mainly on finding information but doesn't mention specific search criteria used or explain how information relates to questions or problem.	Does a good job of explaining the strategy behind the work of investigators. The analysis of research materials or data are described.	Focuses on how research materials or data were used to pursue specific questions or problems. Explains why the actions of the investigators were expected to lead to a resolution of the question or a solution to the problem.
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Unacceptable/accepted/Exemplary

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Department of Computer Science & Engineering

Rubrics for Evaluating the Projects

Academic Year/Semester:

Name and Reg. No. of Batch mates:

Title of the Paper:

Review Number/Date of Review:

Evaluated by:

TRAIT	Marks	Excellent	Good	Average	Score (20)
Understanding background and topic	2	Well formulated	Formulated	The problem is not well formulated	
Identifying the requirements of the proposed work	2	Well identified	Identified	Identified but not to the proposed work	
Formulating the problem with domain knowledge	2	Well designed	Designed	Not well designed	
Attending reviews regularly and showing the progress	2	regularly	Not much regular	irregular	
Capacity to work effectively in tandem with team members	2	Effective participation	Shown participation	Less effective participation	
Capability to communicate and present effectively in reviews	2	Effectively presented	Presented	Not effectively presented	
Summaries, algorithms and highlights the project features	4	Well designed	Designed	Not well designed	
Question and Answer	4	Answered all	Answered some	Didn't answer properly	

Review 1: 20marks

Grade secured: Excellent/Good/Satisfactory/Unsatisfactory

S.A Engineering College, Chennai -77
Department of Computer Science & Engineering

Rubrics for Evaluating the Projects

Academic Year/Semester:

Name and Reg. No. of Batch mates:

Title of the Paper:

Review Number/Date of Review:

Evaluated by:

TRAIT	Marks	Excellent	Good	Average	Score (30)
Updating the existing knowledge for extension of recent works	3	Well formulated	Formulated	The problem is not well formulated	
Ability to frame the design for proposed output	3	Well designed	Designed	Not well designed	
Attending reviews regularly and showing the progress in the satisfactory level	2	regularly	Not much regular	Irregular	
Capacity to work effectively in tandem with teams members	2	Effective participation	Shown participation	Less effective participation	
Capability to communicate and present effectively in Reviews	2	Effectively presented	Presented	Not effectively presented	
Summaries, algorithms and highlights the project features	4	Well designed	Designed	Not well designed	
Question and Answer	4	Answered all	Answered some	Didn't answer properly	
Implementation	10	Implemented well	Implemented some	Dint implement properly	

Review 2: 30marks

Grade secured: Excellent/Good/Satisfactory/Unsatisfactory

S.A Engineering College, Chennai -77
Department of Computer Science & Engineering

Rubrics for Evaluating the Projects

Academic Year/Semester:

Name and Reg. No. of Batch mates:

Title of the Paper:

Review Number/Date of Review:

Evaluated by:

TRAIT	Marks	Excellent	Good	Average	Score (50)
Ability to frame the design for proposed output	2	Well formulated	Formulated	The problem is not well formulated	
Ability to develop the software for proposed work	4	Well designed	Designed	Not well designed	
Attending reviews regularly and showing the progress in the satisfactory level	2	regularly	Not much regular	Irregular	
Capacity to work effectively in tandem with teams members	2	Effective participation	Shown participation	Less effective participation	
Capability to communicate and present effectively in Reviews	2	Effectively presented	Presented	Not effectively presented	
Preparation of quality Reports	10	Well designed	Designed	Not well designed	
Summaries, algorithms and highlights the project features	4	Well designed	Designed	Not well designed	
Question and Answer	4	Answered all	Answered some	Didn't answer properly	
Implementation	20	Implemented well	Implemented some	Not implement properly	

Review 3: 50marks

Grade secured: Excellent/Good/Satisfactory/Unsatisfactory

Project Review Assessment Criteria (PRAC)

Criteria	Description
C1	Understanding background and topic
C2	Identifying the requirements of the proposed work
C3	Updating the existing knowledge for extension of recent works
C4	Formulating the problem with domain knowledge
C5	Ability to frame the design for proposed output
C6	Ability to develop the software for proposed work
C7	Attending reviews regularly and showing the progress
C8	Capacity to work effectively in tandem with team members
C9	Capability to communicate and present effectively in reviews
C10	Preparation of quality Reports
C11	Summaries, algorithms and highlights the project features
C12	Question and Answer
C13	Implementation

Project Review Assessment Criteria (PRAC) Rubrics

REVIEW	MARKS
FIRST	20
SECOND	30
THIRD	50

FIRST REVIEW (20 Marks)

Criteria	Description	Marks
C1	Understanding background and topic	2
C2	Identifying the requirements of the proposed work	2
C4	Formulating the problem with domain knowledge	2
C7	Attending reviews regularly and showing the progress	2
C8	Capacity to work effectively in tandem with team members	2
C9	Capability to communicate and present effectively in reviews	2
C11	Summaries, algorithms and highlights the project features	4
C12	Question and Answer	4
	Total Marks	20

SECOND REVIEW (30 Marks)

Criteria	Description	Marks
C3	Updating the existing knowledge for extension of recent works	3
C5	Ability to frame the design for proposed output	3
C7	Attending reviews regularly and showing the progress in the satisfactory level	2
C8	Capacity to work effectively in tandem with teams members	2
C9	Capability to communicate and present effectively in Reviews	2
C11	Summaries, algorithms and highlights the project features	4
C12	Question and Answer	4
C13	Implementation	10
	Total Marks	30

THIRD REVIEW (50 Marks)

Criteria	Description	Marks
C5	Ability to frame the design for proposed output	2
C6	Ability to develop the software for proposed work	4
C7	Attending reviews regularly and showing the progress in the satisfactory level	2
C8	Capacity to work effectively in tandem with teams members	2
C9	Capability to communicate and present effectively in Reviews	2
C10	Preparation of quality Reports	10
C11	Summaries, algorithms and highlights the project features	4
C12	Question and Answer	4
C13	Implementation	20
	Total Marks	50

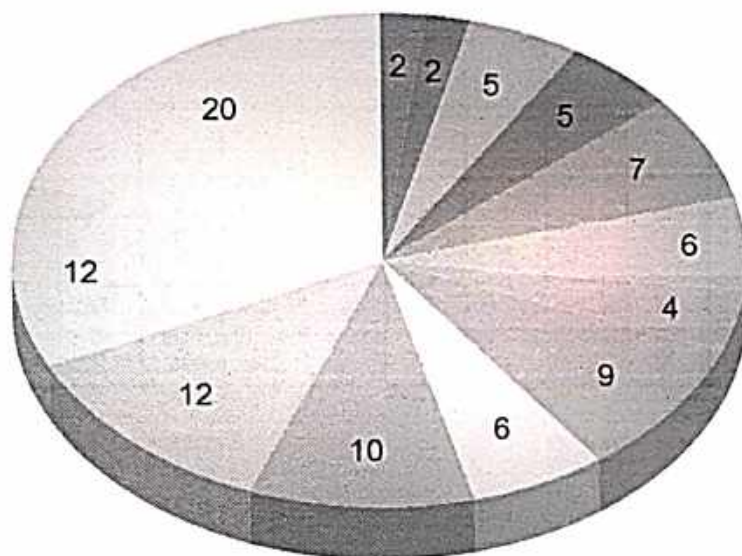
COURSE OUTCOMES

CO's	Description	Criteria
CO1	Formulate a real world challenging problem ethically.	C1
CO2	Develop the requirements of the project and analyse extensively the literature survey.	C2,C3
CO3	Express technical ideas, strategies and methodologies in written and oral presentations.	C4,C10,C7,C8
CO4	Develop sustainable solutions by formulating proper methodology and implementing them.	C5,C13,C9
CO5	Self-learn new tools, algorithms, and/or techniques that contribute to the software solution of the project.	C6,C11,C12

CO PO MAPPING

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO1	PSO2
CS1801.1	3	3	3	3	3	3	3	3	3	3	2	3	3	3
CS1801.2	3	3	2	2	3	3	2	3	2	3	3	3	3	3
CS1801.3	3	3	3	2	2	2	3	2	3	3	2	2	3	3
CS1801.4	3	3	3	2	3	3	3	2	2	1	3	2	3	3
CS1801.5	3	3	2	1	3	2	2	1	2	2	2	2	3	3

FLOW CHART



- criteria 1
- criteria 2
- criteria 3
- criteria 4
- criteria 5
- criteria 6
- criteria 7
- criteria 8
- criteria 9
- criteria 10
- criteria 11
- criteria 12
- criteria 13

COURSE OUTCOMES AND ASSESSMENT METHODS

COURSE OUTCOMES	Contribution	ASSESSMENT METHOD			
		DIRECT METHOD		INDIRECT METHOD	UNIVERSITY RESULTS
		Assessment Tool	Contribution	Tool & Contribution Course End Survey	Contribution
CO1	20%	REVIEWS	17%	3%	80%
CO2					
CO3					
CO4					
CO5					

UNIVERSITY CONTRIBUTION (80%)

Target level	Attainment level	Number of Students
*1	1	N1
**2	2	N2
***3	3	N3

* NUMBER OF STUDENTS WHO SECURED B Grade.

** NUMBER OF STUDENTS WHO SECURED B+ & A Grade.

*** NUMBER OF STUDENTS WHO SECURED A+ & O Grade.

$$((* \times (1/3) + ** \times (2/3) + *** \times (3/3)) / \text{Number of passed Students}) \times (20/100)$$

ATTAINMENT PO THROUGH CO

$$\text{PO Attainment through CO} = [\text{CO1} \times (\text{Level of mapping}/3) + \text{CO2} \times (\text{Level of mapping}/3) + \dots + \text{CO6} \times (\text{Level of mapping}/3)] / 6$$

ATTAINMENT PSO THROUGH CO

$$\text{PSO Attainment through CO} = [\text{CO1} \times (\text{Level of mapping}/3) + \text{CO2} \times (\text{Level of mapping}/3) + \dots + \text{CO6} \times (\text{Level of mapping}/3)] / 6$$

Assessment Tools	Evaluation	Frequency
Internal Assessments	Direct Assessment The department has a structured approach to internal assessments to monitor and enhance student learning throughout the semester. Here's a detailed explanation of the process: Internal Assessments: Internal Assessment I (50 Marks): This test covers Unit 1 and Unit 2 and lasts for 1.30 hours. Internal Assessment II (50 Marks): This test covers Unit 3 and Unit 4 and also lasts for 1.30 hours. Internal Assessment III (100 Marks): This examination covers all units (Unit 1 to Unit 5) and lasts for 3 hours. Attendance for these internal assessments is mandatory. This ensures that all students participate and receive an evaluation of their understanding of the course material. Purpose of Internal Assessments: Performance Evaluation: The results from these tests help faculty gauge the level of knowledge and understanding students have gained. Teaching Methodology Adjustment: Based on the performance data, faculty can identify areas where students may be struggling and adjust their teaching methods accordingly to improve learning outcomes. Structure and Coverage: - The internal assessments are structured to progressively cover the entire syllabus. The Internal Assessment I and II focusing on specific units and the Internal Assessment III covering all units comprehensively. This systematic approach to internal assessments ensures that both students and faculty have a clear understanding of the learning progress and can make necessary adjustments to enhance the educational experience.	Each assessment once in a semester (IAI, IAI, IAI)
End Semester Examination	Will be conducted as per schedule	Once in a semester
Practical Examination	Each student is assigned a system for laboratory work, w/20 marks allocated per experiment as per the syllabus. These marks assess preparation, execution, observation, analysis, and	Once in a semester

	record writing. Continuous assessment through regular experiments ensures students receive ongoing feedback and improve their performance. This structured approach helps achieve learning outcomes and prepares students for real-world applications.	
Project Evaluation	Student projects undergo a thorough evaluation process, assessed periodically through departmental reviews. To comprehensively gauge the students' skills and abilities in project work, three distinct reviews are conducted throughout the project's lifecycle. These reviews serve as milestones, providing opportunities to assess progress, identify areas for improvement, and ensure the project's successful completion.	Once in Final year
Assignments	Assignments play a crucial role in supporting and reinforcing traditional classroom teaching. They provide students with opportunities to further practice and solidify the concepts they've learned in class. This additional practice not only deepens their understanding but also directly contributes to improved performance on assessment tests and examinations. Furthermore, the process of completing assignments fosters and enhances the students' self-learning capabilities, encouraging them to independently explore, research, and apply their knowledge.	Minimum of three assignments per subject
Indirect Assessment		
Course End Survey	Faculty members are actively encouraged to gather feedback from students regarding various aspects of their teaching. This feedback focuses on key areas such as the instructor's clarity in discussing and presenting course material, the effectiveness of instructional examples used, the alignment of assignments and exams with course objectives, and the instructor's enthusiasm for teaching the course. The collected feedback is then carefully analyzed by the faculty member themselves, the group coordinator, and the Head of the Department. Based on the student responses and feedback received, appropriate action is taken to address any concerns or areas for improvement. This feedback mechanism provides valuable insights into the factors that influence student learning, allowing for adjustments in teaching strategies and course content to enhance the overall learning experience.	At the end of semester

Table 3.2.1a Evaluation of Course Outcome

- While university examinations assess student performance using a marks-based system, the final reporting of a candidate's performance in each course will utilize letter grades. These letter grades, each associated with a specific number of points, will be assigned based on the candidate's total marks (out of 100) achieved in each subject. This means that although the raw scores contribute to the final grade calculation, the official record will reflect letter grades, providing a standardized and easily interpretable measure of the student's overall performance in the course. The detailed breakdown of how marks translate into letter grades and their corresponding point values will be provided separately.
- Students will receive a statement of marks that adheres to international standards. This statement will not only list the marks obtained in each subject but also include the Grade Point Average (GPA) and Cumulative Grade Point Average (CGPA). The GPA represents the average of the grade points earned by a student in a given semester, while the CGPA reflects the overall performance across all semesters completed thus far. Including GPA and CGPA on the statement of marks provides a comprehensive and internationally recognized summary of the student's academic progress.

GRADE POINTS AND ITS DESCRIPTION		
Mark Range	Grade Points	Letter Grade
91-100	10	O
81-90	9	A+
71-80	8	A
61-70	7	B+
50-60	6	B
<50	0	RA
-	0	AB
-	-	W

Table 3.2.1b Grade Points Description

Mark Allocation for CO assessment

CO Attainment	Weightage	Assessment Tools
CO attainment	100%	Internal Assessment [I, II, III] and Assignments [20%]
		End Semester Examination [80%]

Table 3.2.1c Calculation of CO attainment

Each individual Course Outcome (CO) is linked to a specific correlation level and assessed using designated evaluation tools. The actual achievement of each CO is measured and compared against a pre-established target attainment level. CO is considered successfully attained if its measured attainment value meets or exceeds this predetermined target.

Theory Course Evaluation

Assessment Tool	Maximum Marks	Marks Scaled to	Weightage
Assignments	50	20%	20%
Internal Assessment – I	50	80%	
Internal Assessment – II	50		
Internal Assessment – III	100		
End Semester Examination	100	80%	80%

Table 3.2.1d Distribution of marks for Theory Course Evaluation

Laboratory Evaluation

Component of Evaluation	Internal/ External	Marks	Marks Scaled to	Weightage
Laboratory Experiments	Internal	20 marks per experiment	20%	20%
Model Laboratory Examination		100	20%	
End Semester Examination	External	100	80%	80%
Laboratory: Total				100%

Table 3.2.1e Distribution of marks for laboratory Evaluation

Theory + Integrated Laboratory Evaluation

Component of Evaluation	Internal/ External	Marks	Marks Scaled to	Weightage
Assignments	Internal	50	20%	20%
Internal Assessment – I		50	80%	
Internal Assessment – II		50		
Internal Assessment – III		100		
Model Laboratory Examination		100	20%	
End Semester Examination	External	100	80%	80%
Theory + Integrated Laboratory: Total				100%

Table 3.2.1f Distribution of marks for Theory + Integrated Laboratory evaluation

Project Evaluation

Component of Evaluation	Internal/ External	Marks	Marks Scaled to	Weightage
Review-1	Internal	20	20%	20%
Review-2		30		
Review-3		50		
Viva-Voce Examination	External	100	80%	80%
Project: Total				100%

Table 3.2.1g Distribution of marks for project evaluation

Rubrics for Project Review

Rubrics for Project Assessment

Rubrics for project assessment are structured scoring guides designed to evaluate student performance on projects based on predefined criteria and standards. They provide a clear and consistent framework for assessing various aspects of a project, such as research quality,

creativity, technical skills, presentation, and adherence to guidelines. Each criterion is broken down into specific performance levels, ranging from excellent to poor, with detailed descriptors for each level. This ensures transparency and fairness in grading, helping students understand expectations and areas for improvement. Rubrics also streamline the assessment process for educators, enabling them to provide constructive feedback and measure learning outcomes effectively. By using rubrics, institutions can maintain academic rigor, enhance student engagement, and ensure alignment with course and program objectives.

Frist Review:

TRAIT	Marks	Excellent	Good	Average	Score (20)
Understanding background and topic	2	Well formulated	Formulated	The problem is not well formulated	
Identifying the requirements of the proposed work	2	Well identified	Identified	Identified but not to the proposed work	
Formulating the problem with domain knowledge	2	Well designed	Designed	Not well designed	
Attending reviews regularly and showing the progress	2	regularly	Not much regular	irregular	
Capacity to work effectively in tandem with team members	2	Effective participation	Shown participation	Less effective participation	
Capability to communicate and present effectively in reviews	2	Effectively presented	Presented	Not effectively presented	
Summaries, algorithms and highlights the project features	4	Well designed	Designed	Not well designed	
Question and Answer	4	Answered all	Answered some	Didn't answer properly	

Table 3.2.1h Rubrics for first review

Second Review:

TRAIT	Marks	Excellent	Good	Average	Score (30)
Updating the existing knowledge for extension of recent works	3	Well formulated	Formulated	The problem is not well formulated	
Ability to frame the design for proposed output	3	Well designed	Designed	Not well designed	
Attending reviews regularly and showing the progress in the satisfactory level	2	regularly	Not much regular	Irregular	
Capacity to work effectively in tandem with teams members	2	Effective participation	Shown participation	Less effective participation	
Capability to communicate and present effectively in Reviews	2	Effectively presented	Presented	Not effectively presented	

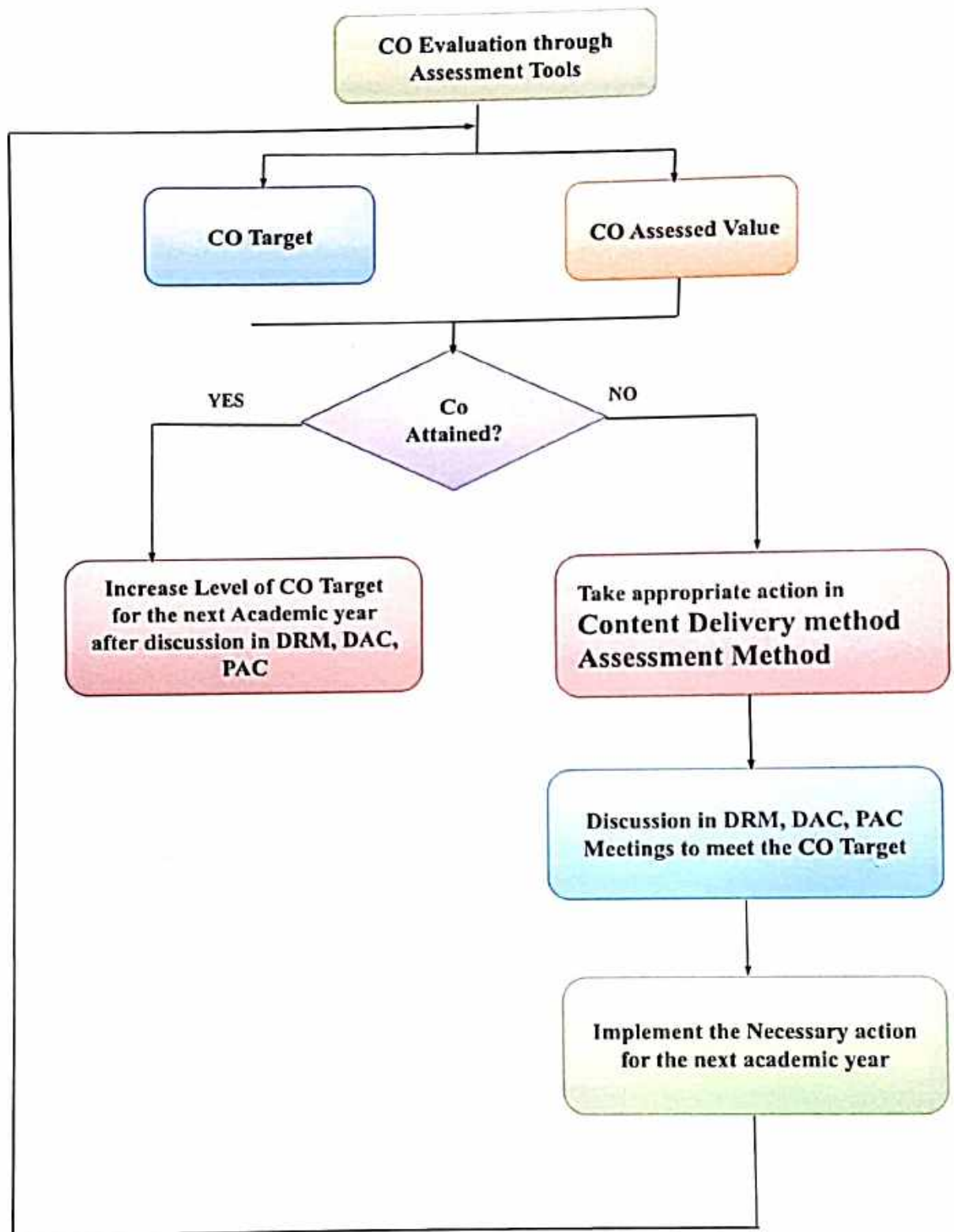
Summaries, algorithms and highlights the project features	4	Well designed	Designed	Not well designed	
Question and Answer	4	Answered all	Answered some	Didn't answer properly	
Implementation	10	Implemented well	Implemented some	Dint implement properly	

Table 3.2.1i Rubrics for second review

Third Review:

TRAIT	Marks	Excellent	Good	Average	Score (50)
Ability to frame the design for proposed output	2	Well formulated	Formulated	The problem is not well formulated	
Ability to develop the software for proposed work	4	Well designed	Designed	Not well designed	
Attending reviews regularly and showing the progress in the satisfactory level	2	regularly	Not much regular	Irregular	
Capacity to work effectively in tandem with teams members	2	Effective participation	Shown participation	Less effective participation	
Capability to communicate and present effectively in Reviews	2	Effectively presented	Presented	Not effectively presented	
Preparation of quality Reports	10	Well designed	Designed	Not well designed	
Summaries, algorithms and highlights the project features	4	Well designed	Designed	Not well designed	
Question and Answer	4	Answered all	Answered some	Didn't answer properly	
Implementation	20	Implemented well	Implemented some	Not implement properly	

Table 3.2.1j Rubrics for third review



Attainment levels for all theory and practical courses individually

Attainment Levels vs. Targets for End Semester Examination

Level 3: > 75%, Students have scored more than 75% in the End Semester Examination

Level 2: >65% - <=75%, Students have scored more than 65% in the End Semester Examination

Level 1: >50% - <=65%, Students have scored more than 50% in the End Semester Examination

Level 0: <50%, Students have scored less than 50% in the End Semester Examination

Attainment Levels vs. Targets for Internal Assessment (IA-I, IA-II, IA-III)

Level 3: > 75%, Students have scored more than 75% in the Internal Assessment Examination

Level 2: >65% - <=75%, Students have scored more than 65% in the Internal Assessment Examination

Level 1: >50% - <=65%, Students have scored more than 50% in the Internal Assessment Examination

Level 0: <50%, Students have scored less than 50% in the Internal Assessment Examination

Indirect Attainment of CO

In this method, we consider the feedbacks (course end survey) of students on the framed questionnaires at the completion of course. Students are asked to rate course on a scale of 5.

In 3.2.2, attainment of course outcomes of all courses with respect to attainment levels has been recorded.

This attainment of course outcomes involves systematically documenting and analyzing the achievement of Course Outcomes (COs) across all courses within a program, measured against predefined attainment levels. It targets for each CO, which serve as the basis for evaluation. Data is collected from various assessment methods, such as exams, assignments, projects, and practical evaluations, to determine how well students have in levels are then calculated, often expressed as percentages or scores, and compared to the set benchmarks. This record provides a comprehensive overview of student performance, highlighting areas of strength and identifying gaps that need attention.

Course Attainment Calculations

Weightage given to Continuous Internal Assessments = 20%

Weightage given to End Semester Examination = 80%

Total Attainment = Continuous Internal Assessments *0.2 + End Semester Examination *0.8

$$\text{Attainment Percentage} = \frac{\text{Number of Students Scoring above the Threshold}}{\text{Total Number of Students Appearing for that Particular CO}}$$

3.3 ATTAINMENT OF PROGRAM OUTCOMES AND PROGRAM SPECIFIC OUTCOMES

3.3.1 Describe assessment tools and processes used for measuring the attainment of each Program Outcome and Program Specific Outcomes (10)

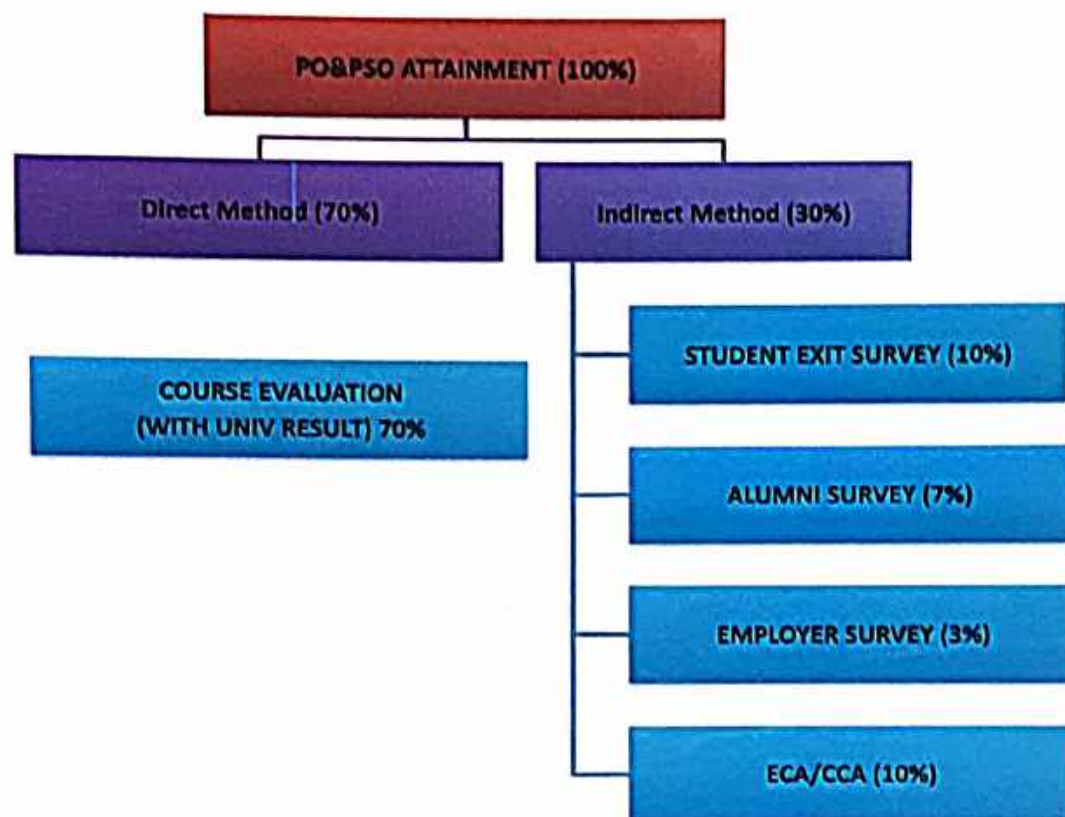
Assessment Process	Evaluation	Frequency
Internal Assessments	The department has a structured approach to internal assessments to monitor and enhance student learning throughout the semester. Here's a detailed explanation of the process: Internal Assessments: Internal Assessment I (50 Marks): This test covers Unit 1 and Unit 2 and lasts for 1.30 hours. Internal Assessment II (50 Marks): This test covers Unit 3 and Unit 4 and also lasts for 1.30 hours. Internal Assessment III (100 Marks): This examination covers all units (Unit 1 to Unit 5) and lasts for 3 hours. Compulsory Attendance: Attendance for these internal assessments is mandatory. This ensures that all students participate and receive an evaluation of their understanding of the course material. Purpose of Internal Assessments: Performance Evaluation: The results from these tests help faculty gauge the level of knowledge and understanding students have gained. Teaching Methodology Adjustment: Based on the performance data, faculty can identify areas where students may be struggling and adjust their teaching methods accordingly to improve learning outcomes. Structure and Coverage:	Each assessment once in a semester

	The internal assessments are structured to progressively cover the entire syllabus. The Internal Assessment I and II focusing on specific units and the Internal Assessment III covering all units comprehensively. This systematic approach to internal assessments ensures that both students and faculty have a clear understanding of the learning progress and can make necessary adjustments to enhance the educational experience. Will be conducted as per schedule	
End Semester Examination		Once in a semester
Practical Examination	Each student is assigned a system for laboratory work, with 20 marks allocated per experiment as per the syllabus. These marks assess preparation, execution, observation, analysis, and report writing. The lab sessions aim to develop practical skills, provide hands-on experience, and enhance problem-solving abilities. Continuous assessment through regular experiments ensures students receive ongoing feedback and improve their performance. This structured approach helps achieve learning outcomes and prepares students for real-world applications.	Once in a semester
Project Evaluation	Student projects undergo a thorough evaluation process, assessed periodically through departmental reviews. To comprehensively gauge the students' skills and abilities in project work, three distinct reviews are conducted throughout the project's lifecycle. These reviews serve as milestones, providing opportunities to assess progress, identify areas for improvement, and ensure the project's successful completion.	Once in Final year
Assignments	Assignments play a crucial role in supporting and reinforcing traditional classroom teaching. They provide students with opportunities to further practice and solidify the concepts they've learned in class. This additional practice not only deepens their understanding but also directly contributes to improved performance on assessment tests and examinations. Furthermore, the process of completing assignments fosters and enhances the students' self-learning capabilities, encouraging them to independently explore, research, and apply their knowledge.	Minimum of three assignments per subject
INDIRECT ASSESSMENT		
EVENT	ASSESSMENT TOOL	

Course End Survey	Faculty members are actively encouraged to gather feedback from students regarding various aspects of their teaching. This feedback focuses on key areas such as the instructor's clarity in discussing and presenting course material, the effectiveness of instructional examples used, the alignment of assignments and exams with course objectives, and the instructor's enthusiasm for teaching the course. The collected feedback is then carefully analyzed by the faculty member themselves, the group coordinator, and the Head of the Department. Based on the student responses and feedback received, appropriate action is taken to address any concerns or areas for improvement. This feedback mechanism provides valuable insights into the factors that influence student learning, allowing for adjustments in teaching strategies and course content to enhance the overall learning experience.	Course End Survey
Alumni Survey	An alumni survey is a tool used by educational institutions to gather feedback from former students about their experiences, career progress, and the impact of their education on their professional and personal lives. This survey typically includes questions about employment status, job satisfaction, skills acquired during their studies, and suggestions for improving the curriculum or campus facilities. The insights gained from alumni surveys help institutions assess the effectiveness of their programs, identify areas for improvement, and strengthen connections with their graduates. By understanding alumni perspectives, institutions can make informed decisions to enhance the quality of education, better prepare future students for the workforce, and maintain a strong, engaged alumni network.	Alumni Survey
Student Exit Survey	A student exit survey is a feedback mechanism conducted by educational institutions to gather insights from students who are about to graduate or leave the program. This survey typically covers aspects such as their overall academic experience, quality of teaching, campus facilities, support services, and preparedness for future career or educational pursuits. The purpose of the survey is to identify strengths and weaknesses in the institution's offerings, understand student satisfaction levels, and pinpoint areas for improvement. The feedback collected helps institutions refine their programs, enhance student support systems, and ensure that future students have a more fulfilling and effective educational experience. It also serves as a valuable tool for maintaining accountability and fostering continuous improvement within the institution.	Student Exit Survey

Employer survey	An employer survey is a feedback tool used by educational institutions or organizations to gather insights from employers about the performance, skills, and competencies of graduates or employees they have hired. This survey typically assesses areas such as job readiness, technical knowledge, soft skills, and overall satisfaction with the hires. The feedback helps institutions evaluate the effectiveness of their academic programs, identify gaps in skills or training, and align their curriculum with industry needs. For organizations, it provides valuable information to improve hiring processes, training programs, and workforce development strategies. By understanding employer perspectives, educational institutions and businesses can enhance collaboration, ensure graduates meet market demands, and foster stronger relationships between academia and industry.	Employer survey
ECA-CCA EVENTS	A department event survey is a feedback tool used by academic departments or organizations to evaluate the success and impact of events such as professional bodies/student chapters/ workshops/seminars/ conferences/ paper presentations /internships /industry visits etc. This survey typically collects participants' opinions on various aspects, including the event's organization, content quality, relevance, speaker effectiveness, and overall experience. The feedback helps identify strengths and areas for improvement, ensuring that future events are more engaging, informative, and aligned with the participants' needs and expectations. By analyzing survey responses, departments can enhance event planning, foster greater participation, and achieve their objectives more effectively, whether they are educational, networking, or community-building goals.	Event Survey

Table 3.3.1a Program Outcomes and Program Specific Outcomes



The PO Assessment Tool is a comprehensive framework designed to evaluate the attainment of Program Outcomes (POs) in an academic program. It includes a variety of assessment methods, such as course evaluations, capstone projects, internships, surveys, and standardized tests, which are aligned with the broader goals and objectives of the program. This tool ensures that the skills, knowledge, and competencies expected at the program level are systematically measured and analyzed. It often incorporates rubrics, performance indicators, and benchmarking criteria to provide a consistent and objective evaluation process.

PO / PSO Assessment Tools:

PO / PSO assessment is done by giving 70% weightage to direct assessment and 30% weightage to indirect assessment. Direct assessment is based on overall CO attainment and CO-PO/PSO mapping. Indirect assessment is done through program exit survey, alumni survey and employer survey. Student exit survey is given a weightage of 10% each and employer survey is given a weightage of 3% each, alumni survey is given a weightage of 7% and events 10%

Quality / Relevance of Assessment Tools And Processes

(i) Direct Assessment Tools and Process:

Direct CO Assessment tools are used for the direct assessment of POs and PSOs. The attainment of each PO corresponding to a particular course is determined from the attainment values obtained for each course outcome related to that PO and the CO-PO mapping values. Similarly, the values of PSO attainment are also determined.

(ii) Indirect Assessment Tools and process:

Indirect assessment is done through student exit survey, alumni survey and employer survey where Student exit survey are given a weightage of 10% each and employer survey are given a weightage of 3% each and alumni survey is given a weightage of 7% and Events 10%.

Process of PO & PSO Attainment

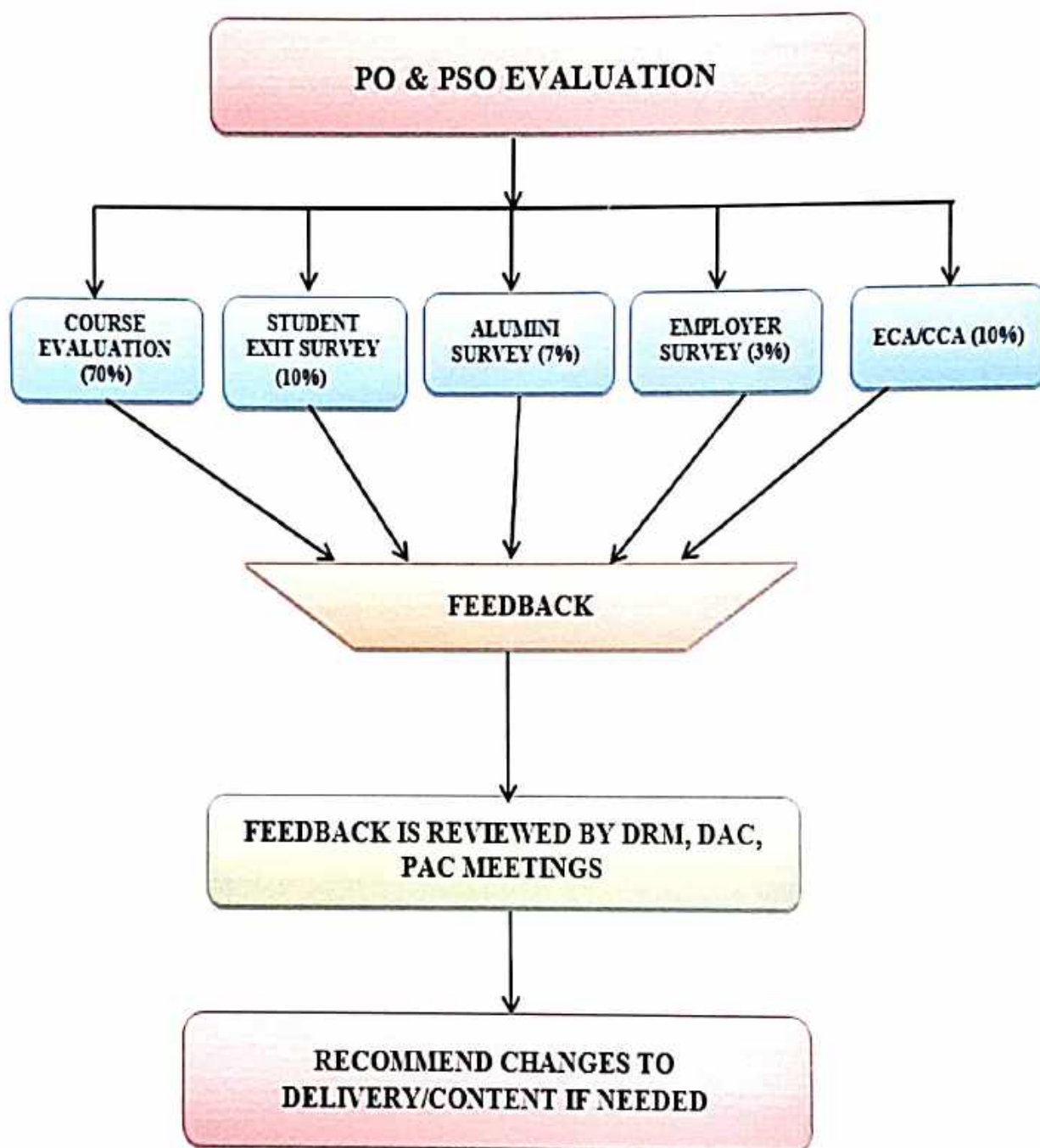
The process of Program Outcome (PO) and Program Specific Outcome (PSO) attainment involves systematically assessing and evaluating the extent to which students achieve the defined learning objectives of a program. It typically begins with the establishment of clear, measurable POs and PSOs aligned with program goals and industry standards. Direct assessment methods, such as exams,

INDIRECT ATTAINMENT

Survey Forms - Questionnaires

Survey forms and questionnaires are essential tools for indirect attainment, which refers to the process of gathering data on learning outcomes or competencies through indirect measures rather than direct assessments like exams or performance evaluations. These tools allow educators, researchers, or organizations to collect feedback, self-reported data, or perceptions from students, employees, or participants about their knowledge, skills, attitudes, or experiences. Indirect attainment provides valuable insights into areas such as program effectiveness, student satisfaction, or skill development, complementing direct measures to offer a more comprehensive understanding of outcomes. By analyzing responses from surveys or questionnaires, stakeholders can identify strengths, weaknesses, and areas for improvement, ultimately supporting decision-making and continuous improvement efforts.

ASSESSMENT METHODS TO ASSESS THE IMPACT OF DELIVERY



S.A.ENGINEERING COLLEGE, CHENNAI – 77
 (An Autonomous Institution, Affiliated to Anna University)
DEPARTMENT OF _____

Student Exit Survey

The purpose of this survey is to obtain Student input on the quality of education, they received and the level of preparation they had in the undergraduate program of Information Technology. The purpose of this survey is to assess the quality of the academic program. We seek your help in completing this survey.

Student Information

1. Name _____
2. Register Number _____
3. Year of graduation _____

Rate the following:

5: Excellent

4: Very good

3: Good

2: Fair

1: Poor

1.	Rate the quality of the courses under taken in the Information Technology department as	
2.	Rate the quality of guidelines followed in the program	
3.	Rate the overall satisfaction with the programme	
4.	Can you apply basic knowledge of computing and mathematics in information technology	
5.	At what level the programme help you to analyze a problem, identify and define the computing requirements appropriate to its solution	
6.	Can you analyze , design and implement the solution for computer-based system using the modern engineering tools	
7.	At what level the programme help you to indulge in research methods for identifying the improvement in the processes	
8.	Can you analyze the impacts of solutions for maintaining sustainability	
9.	Does the programme created an awareness of professional ethics and responsibilities to be applied in real situations.	
10.	Does the programme provided a sufficient foundation for you to possess a good leadership quality	
11.	How well the programme upgraded your communication skills	
12.	Does the programme nurture you to address the social issues	
13.	Rate the level at which the programme helped you to effectively manage projects in diverse environments.	
14.	Does the programme established a strong base for life-long learning.	
15.	How does the programme helped you to study the advanced concepts in core Information Technology	

Date:
Place:

Send to:
Head of the department

Relation of POs and PSOs with questionnaire:

POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
Questions	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12

PSOs	PSO1	PSO2
Questions	Q13	Q14

Evaluation Process:

The questionnaire consists of 15 questions which are relevant for assessing each PO and PSO. The first 12 questions correspond to the 12 POs and the remaining 3 questions are for PSOs (Questions 13 are used to evaluate PSO 1, Question 14 is used to evaluate PSO 2 and Question 15 is used to evaluate PSO 3). Each question has options, namely, Excellent, Very good, Good, Fair, Poor, which is given marks 5,4,3,2 and 1 respectively. The survey results are tabulated and the average values corresponding to each PO and PSO are calculated.

S.A.ENGINEERING COLLEGE, CHENNAI – 77

(An Autonomous Institution, Affiliated to Anna University)

DEPARTMENT OF _____

Alumni Survey Form

Dear Alumni,

Kindly spend your precious time to fill up this Alumni Survey Form.

Provide your valuable inputs about how the Programme you have undergone here helped you to excel in your career which will enable us to know the attainment of our Programme Outcomes and Programme Educational Objectives.

Alumni Information

1. Name _____
2. Batch : _____
3. Phone No: _____
4. E mail id: _____
5. Residential Address _____

Professional Details		
Higher Education Details: (Pursuing / Completed)	Employment Details:	Self-Employment Details:
Degree Name: _____ College/University Name: _____ Address: _____ Period of Study : 20__ to 20__	Designation: _____ Company/Industry Name: _____ Address: _____ Official email id: _____	Company/Industry Registration No. _____ Name: _____ Address: _____

a) Are you currently working in your major field of engineering?

☐ Yes ☐ No

b) Have you ever completed any engineering projects/tasks successfully at workplace?

☐ Yes ☐ No

c) Have you ever achieved or been promoted after joining this organization?

☐ Yes ☐ No

If so mention the designation with company name

Sl No	Designation	Name of the Company

d) Do you have an experience of leading a team (Project Leader/ Project Manager)?

☐ Yes ☐ No

e) Do you enjoy enriching your knowledge by pursuing new courses relevant to your work?

☐ Yes ☐ No

f) Do you find that any one of your completed/ongoing projects is societal friendly?

☐ Yes ☐ No

If yes kindly mention the name of the project and organization name.

Sl No	Project Name	Organization Name

g) Did you receive any awards from your organization?

☐ Yes ☐ No

If yes mention the name of the award with year and organization

Sl No	Name of the Award	Year/ Organization Name

h) To meet the current Job requirements, please specify tools / Novel Technologies, you used other than what you have learnt during the Degree

Any other Suggestions:

Signature of the Alumni with date

(b) Relation of POs and PSOs with questionnaire:

POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
Questions	Q1	Q2	Q3	Q4	Q5	Q7	Q6	Q8	Q9	Q10	Q11	Q12

PSOs	PSO1	PSO2
Questions	Q13	Q14

(c) Evaluation Process:

The questionnaire consists of 14 questions which are relevant for assessing each PO and PSO. The first 12 questions are used to evaluate the 12 POs and the remaining 2 questions are for evaluating PSOs (Questions 13 is used to evaluate PSO 1 and PSO 3, Question 14 is used to evaluate PSO 2). Each question has 5 options, 5-Excellent, 4-Very Good, 3-Good, 2-Average, 1-Need to improve. These marks are tabulated and the average value is shown.

S.A.ENGINEERING COLLEGE, CHENNAI – 77
(An Autonomous Institution, Affiliated to Anna University)

DEPARTMENT OF _____
EMPLOYER FEEDBACK FORM (POST PLACEMENT)

NAME OF THE EMPLOYEE:
ALUMNI DEPARTMENT /ROLL NUMBER:
EMPLOYING ORGANIZATION:

Dear Sir,

As part of our post placement feedback/evaluation process we request you the rate the subject alumni of our college and currently working in your esteemed organization on a 10 point rating by sparing your valuable time and send us back the format in hard copy/email as convenient to you and oblige

Director Corporate Affairs & Placement

ASSESSMENT TRAITS	EXCELLEN T 9	SUPERIO R 8	VERY GOO D 7	GOO D 6	AVERAGE 5	POO R 4
Job Knowledge						
Job Performance						
Learning Interest						
Punctuality & Attendance						
Creative Ability						
Communication /Feed Back						
Technical Skills						
Analytical Skills						
Interpersonal Skills						
Initiative & Leadership						
Total						
YOUR VALUABLE SUGGESTION AS INPUTS FOR CURRICULUM/TRAINING OF STUDENTS						
DATE:	SIGNATURE					

(b) Relation of POs and PSOs with questionnaire:

POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
Questions	Q1	Q8	Q2	Q3	Q5	Q9	Q9	Q4	Q7	Q6	Q10	Q4

PSOs	PSO1	PSO2
Questions	Q1	Q2

(c) Evaluation Process:

The questionnaire consists of 10 questions. These questions are relevant for assessing each PO and PSO. If multiple questions satisfy a PO, then their average is taken. A similar procedure is followed for PSOs also. Each question is having 3 options namely, extremely satisfied, satisfied and somewhat satisfied, which is given marks 9,8,7,6,5 and 4 respectively.

STUDENT EXIT SURVEY (2020-2024 Batch)

[illegible]

10	A	111930CS01011	BALAJI V	To work in IT/ITES industry, To work in government/private sector	5	5	4	4	5	4	2	4	5	4	4	4	4	4
11	A	111930CS01013	BALAMURUGA N K	To work in IT/ITES industry	3	3	3	4	4	4	4	4	4	4	3	4	4	4
13	A	111930CS01014	BELRIN ANIS ABRAHAM G	To work in IT/ITES industry, To work in government/private sector, To start own business	5	4	4	5	4	5	2	4	5	5	5	5	4	4
14	A	111930CS01015	BHUVANESHWARI N	To work in government/private sector, To start own business	5	5	5	5	5	5	2	5	5	5	5	5	5	5
15	A	111930CS01015	CATHERINE JENISITHA J	To work in IT/ITES industry, To work in government/private sector, To start own business	4	4	4	4	4	4	4	4	4	4	4	4	4	4
15	A	111930CS01016	DEEBA ADITHYA A	To work in IT/ITES industry, To work in government/private sector, To start own business	5	5	5	5	5	5	5	5	5	5	5	5	5	5
16	A	111930CS01017	DEEPAK SHIVA M	To work in IT/ITES industry, To start own business	4	4	4	4	4	4	4	4	4	4	4	4	4	4
17	A	111930CS01018	DIANUSRI R	To work in IT/ITES industry, To work in government/private sector, To start own business	5	5	5	5	5	5	5	5	5	5	5	5	5	5
18	A	111930CS01019	DIARANI DHARAN J	To work in IT/ITES industry, To start own business	5	5	5	5	5	5	5	5	5	5	5	5	5	5
19	A	111930CS01030	DIARSHINI R	To work in IT/ITES industry, To start own business, To pursue higher studies	4	4	4	4	4	4	4	4	4	4	4	4	4	4
20	A	111930CS01031	DIAYANITHIVASAN S	To work in IT/ITES industry	5	5	5	1	5	5	5	5	5	5	5	5	5	5
31	A	111930CS01033	DIEEPAN RAJ N	To pursue higher studies	5	5	5	5	4	5	4	5	4	4	4	4	5	5

	A	111930CS01046	HARISH SRINIVAS S R	To work in IT/ITeS industry, To pursue higher studies	5	5	5	2	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
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50	A	111930CS01404	KARTHIK.M(04.09.31)	To work in IT/ITeS industry. To pursue higher studies	5	5	5	3	3	4	4	4	3	5	4	5
				To work in IT/ITeS industry. To start own business. To pursue higher studies	5	5	5	3	5	5	5	5	2	5	5	5
51	B	111930CS01410	SUBASHI D(10.10.31)	To work in IT/ITeS industry. To work in government/private sector	5	5	5	4	4	4	5	5	2	4	5	5
				To work in IT/ITeS industry. To start own business. To pursue higher studies	5	5	5	1	5	5	5	5	2	5	4	5
53	B	111930CS01050	KASHIKA T	To work in IT/ITeS industry	5	5	5	1	5	5	5	5	2	5	4	3
54	B	111930CS01051	KASHIRAJAN R KEZHIA	To work in IT/ITeS industry	4	4	4	1	4	4	4	4	4	4	4	4
55	B	111930CS01053	VALARMATHI A	To work in IT/ITeS industry. To start own business	5	5	5	5	5	5	5	5	5	5	5	5
56	B	111930CS01054	KAUSTHUBAN M	To work in IT/ITeS industry	5	5	5	5	5	5	5	5	5	5	5	5
57	B	111930CS01055	KAVIYA DHARSINI S	To start own business	4	4	4	4	4	4	4	4	4	4	4	4
58	B	111930CS01056	VARDHINI B	To work in IT/ITeS industry	3	4	4	5	3	5	4	4	3	4	4	3
59	B	111930CS01057	KEERTIANA A KESAVANAND AN K	To work in IT/ITeS industry	4	4	4	4	4	4	4	4	4	4	4	4
60	B	111930CS01058	KESHAVA PRIYA V	To work in IT/ITeS industry	4	3	3	4	3	3	3	3	3	3	4	3
61	B	111930CS01059	KISHORE K	To work in IT/ITeS industry. To work in government/private sector. To start own business	4	4	4	4	4	4	4	4	4	4	4	4
63	B	111930CS01060	KRISHNA SWETHA M	To work in IT/ITeS industry	5	5	5	2	5	5	5	5	5	5	5	5
64	B	111930CS01061	LAKITHA R	To work in IT/ITeS industry. To work in government/private sector	5	5	5	5	5	5	5	5	5	5	5	5
65	B	111930CS01063	LALITHSAGAR G	To start own business	4	4	4	2	4	4	4	3	3	3	3	3

[illegible]

C	111930CS01111	SATHISH KUMAR P	To work in IT/ITeS industry, To start own business	3	4	4	4	3	3	3	5	5	4	4	5	5	4
118	C	111930CS01113	SEDHUPATHIP ANDIAN R	5	5	5	4	4	5	5	5	5	5	5	5	5	5
119	C	111930CS01114	SENTHAMIZH SELVAN J	4	4	4	4	3	4	4	4	4	4	4	4	4	5
130	C	111930CS01115	SHARAN S S	4	3	4	4	3	5	4	4	4	5	4	4	4	3
131	C	111930CS01115	SHARNILA P	4	4	4	4	2	4	4	4	4	4	4	4	4	4
133	C	111930CS01116	SHESAGOPAL V	3	3	3	3	3	3	3	3	3	3	3	3	3	3
134	C	111930CS01117	SHIVANI SRIS R	5	5	5	5	3	5	5	5	5	5	5	5	5	5
135	C	111930CS01118	SIDDHARTH UPADHYAYA	4	4	4	4	2	4	4	4	4	4	4	4	4	4
135	C	111930CS01119	SIVA V	3	4	4	4	3	3	3	5	4	4	4	4	4	3
136	C	111930CS01130	SOMNATH C U	4	4	4	4	3	4	4	4	4	4	4	4	4	4
137	C	111930CS01131	SRIBAN M	4	3	3	3	3	3	3	3	3	5	3	4	5	3
138	C	111930CS01133	SRINITHI P	4	4	4	4	4	4	4	4	4	4	4	4	4	4
139	C	111930CS01134	SUBASHI M	5	5	5	5	3	5	5	5	5	5	5	5	5	5
140	C	111930CS01135	SURIYA G	5	5	5	5	3	5	5	5	5	5	5	5	5	5
141	C	111930CS01135	SURYA V	4	4	4	4	1	4	4	4	4	5	3	3	3	3

Sl. No.	Category	Registration No.	Name	To work in IT/ITeS industry, To work in government/private sector	5	4	4	3	4	4	2	4	5	3	5	5	4
144	C	111930CS01137	SURYA D	To work in IT/ITeS industry, To work in government/private sector, To start own business, To pursue higher studies	5	5	5	3	5	5	5	5	5	5	5	5	5
145	C	111930CS01138	SWETHA K	To start own business	4	4	4	1	4	4	2	4	4	4	4	4	4
145	C	111930CS01139	THARUNIKA S	To work in IT/ITeS industry	5	5	5	3	5	5	5	5	5	5	5	5	5
146	C	111930CS01140	UNESH KANNA E	To work in IT/ITeS industry, To start own business, To pursue higher studies	5	5	5	3	5	5	2	5	5	4	5	5	5
147	C	111930CS01141	VARSHA K A	To work in IT/ITeS industry	5	5	5	3	5	5	5	5	5	5	5	5	5
148	C	111930CS01143	VARUN S	To work in IT/ITeS industry	5	5	5	4	4	5	4	5	4	4	4	4	5
149	C	111930CS01144	VARUN V	To work in IT/ITeS industry, To work in government/private sector	5	4	4	3	4	5	1	4	5	5	4	5	5
150	C	111930CS01145	VELIDI HARSHITHA	To work in IT/ITeS industry	5	5	5	2	5	5	2	5	5	5	5	5	5
151	C	111930CS01145	VENKATESAN S	To work in IT/ITeS industry, To work in government/private sector, To pursue higher studies	5	5	5	1	5	4	2	4	4	5	5	5	4
153	C	111930CS01146	VIGNESH L	To work in IT/ITeS industry, To start own business	5	4	5	2	5	5	1	5	5	5	4	5	4
154	C	111930CS01147	VIGNESH S (11.08.3003)	To work in government/private sector, To start own business	5	4	5	2	5	5	2	5	4	5	5	4	4
155	C	111930CS01148	VIGNESH S (13.13.3003)	To work in IT/ITeS industry, To start own business	5	5	5	3	5	5	2	5	5	5	5	5	5
155	C	111930CS01149	VIGNESHSIVIER M	To work in IT/ITeS industry	5	5	5	2	4	5	5	4	4	5	5	5	5

156	C	11190KCS07148	ATISHAL RITHUSH V	To work in IT It's industry. To work in government private sector.	4	4	4	2	4	4	4	4	4	4	4	4	4
157	C	11190KCS07149	VINOD K A M	To work in IT It's industry. To work in government private sector. To start own business. To pursue higher studies.	5	5	5	2	5	5	5	5	5	5	5	5	5
158	C	11190KCS07150	ZIVA SADIK P T	To work in IT It's industry. To work in government private sector. To start own business. To pursue higher studies.	3	4	3	3	3	3	3	3	3	3	3	3	3
159	C	11190KCS07405	MADUMITRA R(14 B 31)	To work in IT It's industry. To work in government private sector. To pursue higher studies.	4	5	4	3	5	3	4	2	5	5	5	4	5
160	C	11190KCS07406	M PRAVEEN KLMAAR(31.103 I)	To work in IT It's industry. To start own business.	3	4	4	4	3	5	5	4	4	5	5	5	4
161	C	11190KCS07407	SASTHA MANIKANDAN S.(97.10.31)	To work in IT It's industry. To work in government private sector. To start own business.	5	5	5	2	5	5	5	5	5	5	5	5	5
162	C	11190KCS07408	VIGNESH M .(05.08.31)	To work in IT It's industry. To work in government private sector.	4	4	4	3	4	4	4	4	4	4	4	4	5

AVERAGE													
STUDENT EXIT SURVEY (100%)													
STUDENT EXIT SURVEY (10%)													
4.49	4.47	4.48	3.46	4.42	4.45	4.45	3.62	4.41	4.51	4.43	4.41	4.47	4.41
89.87	89.47	89.61	69.21	88.42	89.08	89.08	72.37	88.29	90.13	88.55	88.16	89.34	88.29
8.99	8.95	8.96	6.92	8.84	8.91	8.91	7.24	8.83	9.01	8.86	8.82	8.93	8.83
2.70	2.68	2.69	2.08	2.65	2.67	2.67	2.17	2.65	2.70	2.66	2.64	2.68	2.65
Attachment Level													

2020-2024 NATC

[illegible]

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200	201	202	203	204	205	206	207	208	209	210	211	212	213	214	215	216	217	218	219	220	221	222	223	224	225	226	227	228	229	230	231	232	233	234	235	236	237	238	239	240	241	242	243	244	245	246	247	248	249	250	251	252	253	254	255	256	257	258	259	260	261	262	263	264	265	266	267	268	269	270	271	272	273	274	275	276	277	278	279	280	281	282	283	284	285	286	287	288	289	290	291	292	293	294	295	296	297	298	299	300	301	302	303	304	305	306	307	308	309	310	311	312	313	314	315	316	317	318	319	320	321	322	323	324	325	326	327	328	329	330	331	332	333	334	335	336	337	338	339	340	341	342	343	344	345	346	347	348	349	350	351	352	353	354	355	356	357	358	359	360	361	362	363	364	365	366	367	368	369	370	371	372	373	374	375	376	377	378	379	380	381	382	383	384	385	386	387	388	389	390	391	392	393	394	395	396	397	398	399	400	401	402	403	404	405	406	407	408	409	410	411	412	413	414	415	416	417	418	419	420	421	422	423	424	425	426	427	428	429	430	431	432	433	434	435	436	437	438	439	440	441	442	443	444	445	446	447	448	449	450	451	452	453	454	455	456	457	458	459	460	461	462	463	464	465	466	467	468	469	470	471	472	473	474	475	476	477	478	479	480	481	482	483	484	485	486	487	488	489	490	491	492	493	494	495	496	497	498	499	500	501	502	503	504	505	506	507	508	509	510	511	512	513	514	515	516	517	518	519	520	521	522	523	524	5
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MAY 1996 - 2020 2024

S.A. ENGINEERING COLLEGE, CHENNAI 600 077

(An Autonomous Institution, Affiliated to Anna University)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

EVENT SURVEY

2020-2024 BATCH

S.No	EVENTS	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
1	Object Oriented Programming	4.82	4.67	4.71	4.50	4.70	0.00	0.00	0.00	0.00	4.44	0.00	4.50	4.90	4.66
2	Career Awareness and Opportunities in Entrepreneurship	4.75	4.81	4.84	4.41	4.71	0.00	0.00	0.00	0.00	4.43	0.00	4.47	4.78	4.87
3	Guest Lecture on "An industrial overview of database and its administration"	4.73	4.66	4.66	4.42	4.74	0.00	0.00	0.00	0.00	4.40	0.00	4.46	4.94	4.50
4	Guest Lecture on "Use Case Modeling application in Industry"	4.74	4.78	4.68	4.24	4.84	0.00	0.00	0.00	0.00	4.40	0.00	4.46	4.41	4.48
5	Workshop on "Machine Learning & Deep Learning with MATLAB"	4.85	4.65	4.65	4.41	4.80	4.73	4.40	4.53	4.51	4.43	4.36	4.47	4.43	4.53
6	Value Added Course on "Full Stack Development (Client side / Front end-I)"	4.94	4.51	4.67	4.37	4.98	4.71	4.41	4.48	4.46	4.39	4.33	4.46	4.39	4.49
7	Seminar on "Cloud Security Challenges and Opportunities"	4.81	4.51	4.90	4.34	4.48	0.00	0.00	0.00	0.00	4.36	0.00	4.48	4.36	4.51

8	Industrial visit to "Centre for Development of Advanced Computing(CDAC)"	4.69	4.37	4.34	4.39	4.44	4.55	4.38	4.50	4.47	4.40	4.29	4.44	4.37	4.47
9	Seminar on "Data analytics and Linux operating system"	4.72	4.39	4.37	4.33	4.45	4.39	4.43	4.48	4.45	4.41	4.33	4.98	4.40	4.49
10	Guest Lecture on "Internet of Things"	4.45	4.41	4.35	4.37	4.45	0.00	0.00	0.00	0.00	4.41	0.00	4.90	4.41	4.48
11	Guest lecture on "Experimenting Formal Languages using JFLAP"	4.86	4.39	4.73	4.44	4.43	0.00	0.00	0.00	0.00	4.39	0.00	4.65	4.40	4.50
12	Online Workshop on "Demonstration of virtual lab experiments for stake holders"	4.69	4.39	4.63	4.44	4.62	4.54	4.39	4.48	4.43	4.39	4.33	4.81	4.70	4.50
13	Guest Lecture on "Advanced Topics in Computer networks"	4.67	4.42	4.63	4.50	4.84	0.00	0.00	0.00	0.00	4.76	0.00	4.47	4.97	4.54
14	Expert lecture on "CCNA - Cisco Certified Network Associate" (CODING CLUB ACTIVITY)	4.88	4.73	4.72	4.51	4.60	0.00	0.00	0.00	0.00	4.73	0.00	4.50	4.85	4.52
15	PROJECT EXPO	4.68	4.72	4.68	4.65	4.54	4.70	4.46	4.54	4.55	4.82	4.37	4.78	4.82	4.52
16	One day workshop on "Cyber Security"	4.61	4.70	4.74	4.49	4.24	4.83	4.46	4.22	4.55	4.61	4.41	4.70	4.93	4.48
17	Guest Lecture on "Cyber Security- Advanced Persistent Threats".	4.75	4.81	4.78	4.59	4.57	0.00	0.00	0.00	0.00	4.82	0.00	4.71	4.95	4.50

18	Guest Lecture on "Entrepreneurship/startup my story-motivational session by successful entrepreneur/startup founder."	4.74	4.75	4.72	4.53	4.29	0.00	0.00	0.00	4.97	0.00	4.84	4.99	4.34
19	CSI FOUNDATION DAY	4.67	4.80	4.70	4.22	4.31	0.00	0.00	0.00	4.83	0.00	4.72	4.85	4.32
20	Guest Lecture on "Artificial Intelligence and Data Engineering"	4.79	4.72	4.72	4.55	4.45	0.00	0.00	0.00	4.85	0.00	4.88	4.96	4.53
21	Value Added Course on "Full Stack Development (Server side)	4.70	4.57	4.58	4.30	4.43	4.73	4.52	4.34	4.43	4.35	4.80	4.91	4.33
22	Industrial Visit to "ISRO"	4.63	4.64	4.70	4.40	4.50	4.96	4.56	4.34	4.43	4.35	4.84	4.91	4.33
23	Guest Lecture on "Block Chain Technologies & its Future"	4.55	4.54	4.64	4.51	4.53	0.00	0.00	0.00	4.93	0.00	4.77	4.98	4.49
24	Guest Lecture on "Virtual Private Cloud"	4.56	4.72	4.60	4.27	4.84	0.00	0.00	0.00	4.84	0.00	4.75	4.99	4.28
25	Value added Course Cyber Security	4.78	4.79	4.72	4.15	4.78	4.78	4.60	4.34	4.43	4.35	4.81	4.95	4.33
26	Value added Course Cyber Security	4.73	4.65	4.55	4.29	4.78	4.78	4.83	4.55	4.43	4.56	4.93	4.76	4.35
27	Value added Course Cyber Security	4.76	4.68	4.66	4.08	4.58	4.64	4.63	4.69	4.36	4.36	4.83	4.89	4.33

28	Two days Workshop and Hands on Training on " Fabrication of Coir Based Products and Employment opportunities in Coir related products for SE Population"	4.78	4.42	4.78	4.41	4.94	4.94	4.94	4.77	4.91	4.34	4.42	4.75	4.82	4.97	4.53
29	Guest Lecture on " Higher Education Guidance"	4.85	4.44	4.76	4.48	4.76	4.94	0.00	0.00	0.00	0.00	4.74	0.00	4.84	4.63	4.74
30	A One Day Intra-College Hackathon on " Code Craft Challenge 2024" organized by CSE dept in Association with Institution's Innovation Council.	4.74	4.66	4.76	4.45	4.30	4.71	4.46	4.75	4.26	4.63	4.86	4.71	4.90	4.99	
31	A Guest Lecture on " Data Science & Machine Learning	4.59	4.55	4.71	4.43	4.31	0.00	0.00	0.00	0.00	0.00	4.57	0.00	4.86	4.66	4.74

AVERAGE	4.73	4.61	4.67	4.40	4.59	4.72	4.52	4.52	4.52	4.44	4.61	4.44	4.70	4.75	4.51
EVENT survey(100%)	94.52	92.15	93.36	88.04	91.76	94.40	90.40	90.40	90.40	88.80	92.18	88.80	93.96	95.06	90.11
EVENT survey(10%)	9.452	9.215	9.336	8.804	9.176	9.440	9.040	9.040	9.040	8.880	9.218	8.880	9.396	9.506	9.011

EVENT ATTAINMENT	2.84	2.76	2.80	2.64	2.75	2.83	2.71	2.71	2.77	2.66	2.82	2.85	2.70
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S.A ENGINEERING COLLEGE, CHENNAI - 77

(An Autonomous Institution Affiliated to Anna University)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

PO and PSO DIRECT ATTAINMENT

BATCH : 2020-2024

BATCH : 2020-2024					Programme Outcomes													Programme Specific Outcomes	
SLNo	SEMESTER	NBA CODE	Subject Code	Subject Name	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	
1	Semester I	C101	HS1101	Technical English	-	-	-	-	-	1.53	0.96	-	2.11	2.88	1.92	1.34	-	-	
2		C102	MA1101	Calculus and its applications	2.71	2.71	1.81	0.86	-	-	-	-	-	-	-	-	-	-	
3		C103	PH1101	Applied Physics	2.84	2.84	-	-	-	-	-	-	-	-	-	-	-	-	
4		C104	CY1101	Engineering Chemistry	1.98	1.19	1.19	-	-	-	-	-	-	-	-	-	-	-	
5		C105	CS1101	Problem Solving And Python Programming	2.77	2.01	2.01	1.85	1.85	-	-	-	-	-	-	1.85	1.85	1.70	1.46
6		C106	ME1101	Engineering Graphics	2.47	2.86	2.86	2.47	1.14	1.14	2.86	0.95	1.91	2.29	2.29	0.95	1.33	-	-
7		C107	BS1101	Physics and Chemistry Laboratory	1.89	1.13	0.94	0.94	-	-	-	-	-	-	-	-	-	-	-
8		C108	CS1102	Problem Solving And Python Programming Laboratory	2.72	2.72	2.53	2.72	-	-	-	-	-	-	-	-	1.94	1.75	1.17
9	Semester II	C110	HS1201	English for Communication	-	-	-	-	1.93	1.20	-	-	-	1.93	2.90	-	1.45	-	-
10		C111	MA1201	Complex Variables and Transforms	2.83	2.64	1.89	-	-	-	-	-	-	-	-	-	-	-	-
11		C112	PH1201	Materials Science	2.93	2.93	-	-	-	-	-	-	0.98	-	-	-	-	-	-
12		C113	EE1202	Basic Electrical, Electronics and Measurement Engineering	2.89	1.93	1.93	-	-	-	-	-	-	-	-	-	-	-	-
13		C114	CS1201	Programming in C	2.90	1.93	2.13	1.93	-	-	-	-	-	-	-	1.93	1.93	1.93	1.93
14		C115	CS1202	Fundamentals of Computing	2.87	2.87	2.15	-	-	-	-	-	-	-	-	-	1.91	1.44	1.12
15		C116	CS1203	Programming in C Laboratory	2.91	1.94	2.14	1.94	1.29	-	-	-	-	-	-	-	-	1.94	1.94
16		C117	GE1201	Engineering Practices Laboratory	2.90	2.90	2.90	1.93	1.93	1.93	2.90	-	2.90	1.93	2.90	-	0.97	-	-
17		C119	CS1204	Fundamentals of Computing Laboratory	2.93	2.93	2.34	1.95	1.37	1.37	-	-	1.37	1.95	1.95	0.98	1.76	2.15	1.95

39	C305	FC1515	Embedded System	2.66	2.66	2.66	1.47	1.32	-	-	-	1.54	1.33	1.77	1.77	1.59	1.59
40	C306	IT1503	Networks Laboratory	2.30	2.11	1.92	-	2.50	-	-	-	1.73	-	-	1.73	2.30	1.92
41	C307	CS1503	Operating Systems Laboratory	1.87	2.24	1.68	-	-	-	-	-	1.49	1.49	-	1.87	1.31	2.05
42	C308	CS1601	Internet Programming	2.44	2.09	2.27	-	2.27	-	1.14	-	1.63	1.30	1.63	1.45	1.78	1.32
43	C309	CS1602	Compiler Design	1.88	1.88	2.26	-	2.83	-	-	-	-	-	-	2.83	1.88	2.08
44	C310	CS1603	Artificial Intelligence	2.21	2.05	1.52	1.52	-	-	-	-	-	1.37	1.87	1.70	1.54	1.52
45	C311	CS1604	Cryptography and Network Security	2.17	1.50	1.84	2.00	1.98	-	-	-	1.33	1.49	-	1.67	2.34	1.99
46	C312	CS1607	Cyber Security	1.91	2.27	1.91	1.56	1.58	2.31	1.72	1.76	1.05	1.22	1.22	1.77	2.61	2.61
47	C313	CS1608	Software Testing	2.44	2.25	2.44	2.06	-	-	1.32	-	-	-	-	-	1.88	1.32
48	C314	CS1605	Internet Programming Laboratory	2.90	2.13	2.90	1.74	2.90	-	1.16	-	2.90	1.35	2.90	1.93	2.13	1.35
49	C315	CS1606	Security Laboratory	2.89	2.50	2.31	1.92	1.54	-	-	-	-	-	-	1.92	2.50	2.60
50	C316	IS1601	Professional Communication Laboratory	-	-	-	-	2.89	2.89	2.89	2.89	2.89	2.89	2.89	2.89	-	-
51	C401	IT1701	Distributed Cloud Computing	2.47	1.90	2.47	-	2.66	-	1.52	1.90	1.52	1.90	-	1.90	1.90	1.90
52	C402	CS1701	Machine Learning	1.73	1.71	1.71	1.55	0.85	-	-	-	-	-	-	-	0.87	1.55
53	C403	MGT701	Principles of Management	1.63	1.63	1.26	-	-	-	1.82	0.93	0.89	0.89	1.29	1.82	1.82	1.63
54	C404	CS1707	Human Computer Interaction	1.88	1.71	1.88	2.24	2.46	2.07	1.71	0.93	1.13	1.13	1.30	1.52	1.52	1.40
55	C405	CS1708	C# and .Net Programming	1.82	2.01	1.82	2.19	2.19	1.79	-	-	-	-	-	1.82	2.01	1.82
56	C406	CS1711	Full Stack Web Development	1.67	1.67	1.67	-	2.17	1.67	2.00	-	1.67	1.67	1.67	2.50	1.67	1.51
57	C407	CS1713	Blockchain Technologies	1.90	1.90	2.28	1.90	2.27	1.34	1.33	1.71	-	-	-	1.90	1.33	1.71
58	C408	ICS1601	Professional Readiness for Innovation, Employability and Entrepreneurship	2.49	2.49	2.64	2.16	1.44	0.96	0.96	1.44	2.88	2.68	2.30	2.88	2.64	2.16
59	C409	IT1702	Cloud Computing Laboratory	2.34	1.95	2.92	1.95	2.92	-	1.95	-	2.92	1.95	2.92	1.95	1.95	1.95

Semester VI

60		C410	CS1703	Summer Internship	1.92	2.12	2.50	2.50	2.88	2.88	1.93	2.12	2.12	1.68	2.40	2.50	1.54
61	Semester VII	C411	CS1702	Machine Learning Laboratory	1.93	1.93	1.93	0.97	0.97	-	-	-	0.97	-	0.97	0.97	1.93
62		C412	GE1801	Professional Ethics in Engineering	-	-	-	-	1.75	1.72	2.60	1.78	1.14	-	1.59	0.87	-
63		C413	CS1808	UI & UX Design	2.13	1.44	1.63	2.13	-	-	-	-	-	-	-	1.42	1.43
64	Semester VIII	C414	CS1809	Green Computing	2.23	1.86	1.49	0.76	0.74	1.12	-	0.57	0.56	0.56	1.31	1.86	1.86
65		C415	CS1812	Digital Marketing	1.78	1.41	1.59	-	2.67	-	-	2.67	-	-	1.43	1.26	1.24
66		C416	CS1801	Project Work	2.87	2.87	2.49	1.91	2.68	2.49	2.10	2.30	2.29	2.30	2.30	2.87	2.87
AVERAGE					2.4	2.2	2.0	1.9	2.1	1.8	1.8	1.9	1.7	1.8	1.8	1.8	1.8
OUT OF 100%					78.5	72.0	65.6	64.3	68.4	60.9	60.1	63.6	56.2	60.5	59.8	60.0	59.4
out of 70%					54.96	50.37	45.91	44.99	47.91	42.60	42.10	44.53	39.37	42.36	41.88	41.99	41.61
Direct Attainment					2.36	2.16	1.97	1.93	2.05	1.83	1.81	1.91	1.69	1.82	1.79	1.80	1.78

N.B.
Process owner

Hold

S.A ENGINEERING COLLEGE, CHENNAI - 77

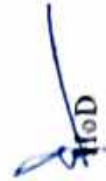
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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

INDIRECT ATTAINMENT (BATCH 2020-2024)

Assessment Methods	Tools	Programme Outcomes												Programme Specific Outcomes	
		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
Indirect Attainment	Student-Exit Survey(10%)	2.70	2.68	2.69	2.08	2.65	2.67	2.67	2.17	2.65	2.70	2.66	2.64	2.68	2.65
	Alumni Survey(7%)	2.67	2.65	2.67	2.62	2.65	2.63	2.65	2.00	2.63	2.68	2.63	2.67	2.63	2.63
	Employer Survey(3%)	2.47	2.50	2.49	2.40	2.46	2.44	2.46	2.47	2.44	2.56	2.43	2.53	2.44	2.44
	ECA - CCA Event (10%)	2.84	2.76	2.80	2.64	2.75	2.83	2.71	2.71	2.66	2.77	2.66	2.82	2.85	2.7
Indirect Attainment		2.67	2.65	2.66	2.44	2.63	2.64	2.62	2.34	2.60	2.68	2.60	2.67	2.65	2.60


Process Owner


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S.A. ENGINEERING COLLEGE, CHENNAI - 77

(An Autonomous Institution Affiliated to Anna University)
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

PO - PSO ATTAINMENT (BATCH 2020-2024)

Assessment Methods	Tools	Programme Outcomes												Programme Specific Outcomes	
		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
Direct Attainment	PO & PSO through CO(70%)	2.36	2.16	1.97	1.93	2.05	2.06	1.83	1.81	1.91	1.69	1.82	1.79	1.80	1.78
InDirect Attainment	PO & PSO through Survey(30%)	2.67	2.65	2.66	2.44	2.63	2.64	2.62	2.34	2.60	2.68	2.60	2.67	2.65	2.60
Final Attainment		2.46	2.32	2.19	1.96	2.20	2.14	1.98	1.89	2.09	1.99	2.00	2.01	2.09	2.04

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Process Owner

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CRITERION 7	Continuous Improvement	75
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7.1 Actions taken based on the results of evaluation of each of the COs, POs & PSOs (30)

Identify the areas of weakness in the program based on the analysis of the evaluation of COs, POs, and PSOs attainment levels. Identify and implement measures to improve POs and PSOs attainment levels for the assessment year, including curriculum interventions, pedagogical initiatives, support system improvements, etc.

POs & PSOs Attainment Levels and Actions for improvement (Batch 2020-24)

POs	Target Level	Attainment Level	Observations
PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.			
PO1	1.98	2.46	Attainment level improved against target level and hence the same method will be continued for the next year.
Action 1: Conduction of tutorial classes for analytical subjects will be followed by next year also. Action 2: More coaching classes will be conducted for slow learners for next academic year. Action 3: Asking viva questions relating to the basic concepts to refresh their fundamentals in laboratory sessions will be continued. Action 4: Motivation of doing internship in reputed companies will be continued. Action 5: Students will be motivated to do NPTEL and other MOOC courses, seminars, debate, concept map, quiz in respective subjects like this. Action 6: Arrangement of guest lectures will be continued on critical topics in relevant subjects to help students in identifying & analysing the real time problems.			
PO2: Problem analysis: Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.			
PO2	1.98	2.32	Attainment level improved against target level and hence same practice will be followed .
Action 1: Relevant activities will be taken to retain the target. Action 2: Tutorial classes for analytical subjects will be conducted like this for next academic year also. Action 3: Conduction of coaching classes for slow learners will be followed. Action 4: Students will be motivated to do NPTEL and other MOOC courses, seminars, debate , concept map, quiz for next academic year also Action 5: Students will be encouraged to participate in external inter college Technical Competitions, coding contests and hackathons like this year for forthcoming academic years.			
PO3: Design/development of solutions: Design solutions for complex engineering problems			

and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations

PO3	1.98	2.19	Attainment level improved against target level and hence the same method will be continued for the next year.
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Action 1: Relevant activities will be taken to retain the target.

Action 2: More tutorial classes for analytical subjects will be conducted like this for next academic year also.

Action 3: Conduction of coaching classes for slow learners will be followed.

Action 4: Conduction of Expert talks, workshop and hands on training session will be continued so that students can understand process of designing and analyzing real life software problems.

Action 4: Students will be motivated to do NPTEL and other MOOC courses, seminars, debate, concept map, quiz for next academic year also

Action 5: Students will be encouraged to participate in external inter college Technical Competitions, coding contests and hackathons like this year for forthcoming academic years.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions

PO4	1.98	1.96	Attainment level decreased against target level
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Action 1: More awareness among students will be created for referring IEEE journal papers /Scopus to enhance their research knowledge, analysis and interpretation of data.

Action 2: Students will be motivated more to take part in national and international conferences for attaining research knowledge.

PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO5	1.98	2.20	Attainment level improved against target level and hence the same method will be continued for the next year
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Action 1: Students exposure to modern tools like Microsoft Azure, Gitlab, AWS Cloud 9, Matlab Apache Net beans etc, will be continued for the next academic year also.

Action 2: Students will be motivated to do NPTEL and other MOOC courses, seminars, debate, concept map, quiz for next academic year also.

Action 3: More motivation by mentors will be done for students to register for Hackathon, Capture the Flag, Paper Presentation, Technical Quiz, Webinars/Seminars conducted by other colleges regarding modern tool usage.

PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO6	1.92	2.14	Attainment level improved against target level and hence same practice will be followed
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- Action 1:** Target will be retained and will be observed for next academic year.
- Action 2:** More exposure of various professional engineering practices followed in the industries for students through industrial visits will be done.
- Action 3:** Activity based learning will be followed for next academic year also.
- Action 4:** Association with professional bodies like CSI, ISTE will be followed to create more awareness among the students about professional engineering practice.
- Action 5:** Continuation of encouragement of students will be done to carry out inter domain projects so that they would realize the importance of a project involving society, safety, health, and the legalities.

PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO7

1.92

1.98

Attainment level improved against target level and hence same practice will be followed

Action 1: Continuation of encouraging students to indulge in mini projects/projects where societal and environmental issues can be addressed will be implemented.

Action 2: Students are strictly instructed to switch off all electrical Equipment /Resources when not in use for all the laboratories.

Action 3: On environmental context Rain water harvesting, Sewage treatment plants, Waste management procedures are employed at our college.

PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

POS

1.92

1.89

Attainment level decreased against target level

Action 1: Professional Ethics will be ensured by giving more importance in scrutinizing the code and project reports of Projects and Mini Projects by doing proper plagiarism check to determine the originality of the project.

PO9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO9

1.92

2.09

Attainment level improved against target level and hence same practice will be followed

Action 1: Attainment level improved against target level and hence same practice will be followed

Action 2: Encouragement of students to participate in various co-curricular and extra-curricular activities in other colleges/sports activities/cultural activities will be continued.

Action 3: Motivating students by mentors to participate in Inter Collegiate Cultural Fest, Hackathon, Technical Quiz, Paper Presentation will be continued..

Action 4: Continuation of the club activities will be done.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO10

1.92

1.99

Attainment level improved against target level and hence same practice will be followed

			Attainment level improved against target level by the subject.
Action 1: Target will be retained and will be observed for next academic year. Action 2: To enhance the employability skills of the students, the training programs on the topics: mock interview, career development, higher studies motivation talk, GATE coaching classes, GATE videos by faculties, entrepreneurship development will be continued next academic year also. Action 3: The Inter/Intra college level project exhibition will be continued for next academic year also.			
PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments			
PO11	1.92	2.00	Attainment level improved against target level and hence same practice will be followed
Action 1: Retention of target will be observed for next academic year. Action 2: Arrangement will be made like this year for students to visit industries on multidisciplinary environments in the next academic year to improve knowledge of the engineering principles. Action 4: Like this year students will be encouraged to write case studies as assignments to know about the recent technologies Action 5: Students will be encouraged to prepare project proposals with the guidance of faculty for government funding agencies like this year.			
PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.			
PO12	1.92	2.01	Attainment level improved against target level and hence same practice will be followed
Action 1: Target attained will be retained Action 2: Mini Projects in curriculum will be continued which helps the students to perform better in placement interviews and higher studies Action 3: Like this year students will be motivated to register for GRE/TOEFL/GATE and other competitive examinations. Action 4: Conduction of GATE coaching classes and GATE coaching videos by faculties will be continued in next academic year also. Action 5: Students will be encouraged to publish papers in various national and International journals/conferences in next academic year also. Action 6: Students will be motivated by mentors to register and get certificates in various online courses on new technologies which would help them in continuous learning in next academic year also.			
PSO1: Apply mathematical foundations, management principles and computational concepts to design and develop efficient solutions for complex technical challenges, adhering to ethical standards and sustainable practices.			
PSO1	1.92	2.09	Attainment level improved against target level and hence same practice will be followed

Action 1: Target attained will be retained Action 2: More demonstrations, case studies, tutorial sessions and quizzes will be adapted into the teaching learning process. Action 3: More ethics related case studies will be given for the knowledge of ethical standards. Action 4: Students will be motivated to do Skill rack, Hackathon, NPTEL courses etc to know efficient solutions for complex technical solutions in next academic year like this year.			
PSO2: Analyse and design computer programs in domains such as algorithms, networking, web development, cloud computing, IoT, machine learning etc., for the development of computer-based systems with varying levels of complexity providing practical, industry-relevant solutions to meet societal and organizational needs			
PSO2	1.92	2.04	Attainment level improved against target level and hence same practice will be followed.
Action 1: Target attained will be retained Action 2: More demonstrations, case studies, tutorial sessions and quizzes will be adapted into the teaching learning process Action 3: Students will be motivated to do Skill rack NPTEL courses etc to know efficient solutions for complex technical solutions in next academic year like this year., Action 4: More guest lectures will be given to students i the industry related technical domains.			

Table 7.1.1 POs & PSOs Attainment Levels and Actions

SA ENGINEERING COLLEGE, CHENNAI-77

DEPARTEMENT OF COMPTON SCIENCE & ENGINEERING

PRO and PNCX AT FALSAHAT

15 JUL 2003

Sl.No	SEMESTER	NBA Code	Subject Code	Subject Name	Programme Outcomes												Programme Specific Outcomes			
					PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	
1	Semester I	C101	HS8151	Communicative English	-	-	-	-	-	-	-	-	-	-	-	-	-	16	10	13
2		C102	MA8151	Engineering Mathematics - I	2.2	2.2	2.3	-	-	-	-	-	-	1.9	2.9	2.9	-	19	16	13
3		C103	PH8151	Engineering Physics - I	2.6	2.6	2.6	-	-	-	-	-	-	-	1.5	-	-	16	19	-
4		C104	CS8151	Engineering Chemistry - I	2.8	1.8	1.8	-	-	-	-	-	-	-	-	-	-	17	14	-
5		C105	GL8151	Problem Solving And Python Programming	2.9	1.9	1.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	10	-	10
6		C106	GL8152	Engineering Graphics	2.4	2.4	1.6	-	1.6	-	-	-	-	2.4	2.4	1.6	-	16	17	17
7		C107	GE8161	Problem Solving And Python Programming Laboratory	2.8	2.8	2.8	-	2.8	-	2.8	-	-	2.8	2.8	1.8	-	18	11	2.7
8	Semester I	C108	HS8161	Physics and Chemistry Laboratory - I	2.9	2.9	2.9	-	-	-	-	-	-	2.9	2.9	2.0	-	-	-	10
9		C109	HS8151	Technical English	-	-	-	-	2.8	-	2.8	2.8	2.8	2.8	2.8	2.8	2.8	1.9	16	19
10		C110	MA8151	Mathematics - II	2.8	2.8	2.8	-	-	-	-	-	-	-	1.9	-	-	19	21	14
11		C111	PH8152	Physics for Information science	2.9	2.9	1.9	-	-	-	-	-	-	-	-	-	-	16	10	13
12		C112	EE8155	Basic Electrical/Electronics and Measurement Engineering	2.5	1.6	1.6	-	-	-	-	-	-	-	-	-	-	11	16	20
13		C113	GL8151	Environmental Science and Engineering	1.1	0.9	1.1	-	-	-	-	-	1.4	0.9	0.9	0.9	-	11	10	10
14		C114	CS8151	Programming in C	3.0	2.6	2.6	-	-	-	-	-	-	1.7	1.7	1.7	-	17	22	10
15	Semester II	C115	GL8151	Engineering Practices Laboratory	2.9	2.9	2.9	2.0	2.0	2.9	2.9	2.9	2.9	2.9	2.9	2.9	2.9	10	-	-
16		C116	CS8151	C programming Laboratory	2.7	2.7	2.7	-	-	-	-	-	-	1.8	1.5	1.5	-	15	20	14
17		C201	MA8151	Discrete Mathematics	2.7	2.9	1.9	-	-	-	-	-	-	-	1.9	-	-	-	2.6	2.3
18		C202	CS8151	Digital Principles and System Design	2.7	2.9	1.9	-	-	-	-	-	-	-	-	-	-	10	19	19
19		C203	CS8151	Data Structure	2.4	2.4	2.4	-	-	-	-	2.0	-	-	2.0	-	-	20	19	12
20		C204	CS8152	Object Oriented Programming	2.9	2.1	2.7	-	-	-	-	-	-	-	2.0	-	-	20	17	16
21		C205	EE8155	Communication Engineering	2.8	2.8	2.7	-	-	-	-	-	-	-	-	-	3.0	2.9	10	
22	Semester III	C206	CS8151	Data Structure Laboratory	2.8	2.8	2.8	-	-	-	-	-	-	-	2.1	1.7	-	21	2.5	1.7
23		C207	CS8153	Object Oriented Programming Laboratory	2.7	2.3	2.1	-	-	-	-	-	-	2.0	2.3	1.8	-	23	11	2.7
24		C208	CS8152	Digital System Laboratory	2.7	2.4	2.7	-	-	-	-	1.8	-	1.9	1.2	1.2	-	14	24	20
25		C209	HS8151	Interpersonal Skills/Listening & Speaking	-	-	-	-	-	-	-	-	-	2.7	1.4	2.7	-	19	10	20
26		C210	MA8152	Probability and Queuing Theory	1.4	1.4	0.8	-	-	-	-	-	-	-	1.1	0.8	-	0.9	-	2.0
27		C211	CS8151	Computer Architecture	2.6	2.7	1.8	-	-	-	-	-	-	-	-	-	-	10	11	0.9
28		C212	CS8151	Computer Architecture	2.6	2.7	1.8	-	-	-	-	-	-	-	-	-	-	10	11	0.9

[illegible]

60	C406	IT8075	Software Project Management	28	20	21	-	-	23	-	24	24	23	25	24	21	21	23
61	C407	CS8073	C++ and Net Programming	18	-	26	-	28	-	-	-	17	15	-	-	21	-	22
62	C408	CS8079	Human Computer Interaction	27	18	18	-	19	-	-	-	-	-	-	-	20	18	-
63	C409	CS8711	Cloud computing Laboratory	29	17	17	25	17	04	-	-	-	-	-	05	19	20	10
64	C410	IT8761	Security Laboratory	29	29	24	-	29	-	-	29	19	-	-	19	29	16	19
65	C411	CE8076	Professional Ethics in Engineering	-	-	-	-	-	14	12	27	09	09	-	28	23	-	22
66	C412	CS8078	Green Computing	26	22	24	-	-	-	-	-	-	-	-	-	19	20	23
67	C413	CS5080	Information Retrieval Technique	24	23	23	-	-	-	-	-	-	-	-	-	20	16	13
68	C414	CS8911	Project work	17	22	19	19	17	16	19	14	19	19	20	17	20	13	14
Semester VIII																		
AVERAGE				24	22	21	18	20	18	16	18	18	18	21	18	18	18	14
OUT OF 70%				56.87	51.40	49.65	41.57	47.26	42.23	37.50	43.08	42.43	42.42	49.39	41.06	42.17	42.35	33.43


 Process Owner

 HOD

CRITERION 7	Continuous Improvement	75
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1. Actions taken based on the results of evaluation of each of the COs, POs & PSOs (30)

Identify the areas of weakness in the program based on the analysis of the evaluation of COs, POs, and PSOs attainment levels. Identify and implement measures to improve POs and PSOs attainment levels for the assessment year, including curriculum interventions, pedagogical initiatives, support system improvements, etc.

POs & PSOs Attainment Levels and Actions for improvement (Batch 2019-23)

POs	Target Level	Attainment Level	Observations
PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.			
PO1	2.01	2.48	Attainment level improved against target level and hence the same method will be continued for the next year.
Action 1: Conduction of tutorial classes for analytical subjects will be followed by next year also. Action 2: More coaching classes will be conducted for slow learners for next academic year. Action 3: Asking viva questions relating to the basic concepts to refresh their fundamentals in laboratory sessions will be continued. Action 4: Motivation of doing internships in reputed companies will be continued. Action 5: Students will be motivated to do NPTEL and other MOOC courses, seminars, debate, concept map, quiz in respective subjects like this. Action 6: Arrangement of guest lectures will be continued on critical topics in relevant subjects to help students in identifying & analysing the real time problems.			
PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.			
PO2	2.01	2.31	Attainment level improved against target level and hence same practice will be followed .
Action 1: Relevant activities will be taken to retain the target. Action 2: Tutorial classes for analytical subjects will be conducted like this for next academic year also. Action 3: Conduction of coaching classes for slow learners will be followed. Action 4: Students will be motivated to do NPTEL and other MOOC courses, seminars, debate , concept map, quiz for next academic year also Action 5: Students will be encouraged to participate in external inter college Technical			

Competitions, coding contests and hackathons like this year for forthcoming academic years.			
PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.			
PO3	2.01	2.26	Attainment level improved against target level and hence the same method will be continued for the next year.
Action 1: Relevant activities will be taken to retain the target. Action 2: More tutorial classes for analytical subjects will be conducted like this for next academic year also. Action 3: Conduction of coaching classes for slow learners will be followed. Action 4: Conduction of Expert talks, workshop and hands on training session will be continued so that students can understand process of designing and analyzing real life software problems. Action 4: Students will be motivated to do NPTEL and other MOOC courses, seminars, debate , concept map, quiz for next academic year also Action 5: Students will be encouraged to participate in external inter college Technical Competitions, coding contests and hackathons like this year for forthcoming academic years.			
PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.			
PO4	2.01	2.02	Attainment level improved against target level and hence same practice will be followed
Action 1: More awareness among students will be created for referring IEEE journal papers /Scopus to enhance their research knowledge, analysis and interpretation of data. Action 2: Students will be motivated more to take part in national and international conferences for attaining research knowledge.			
PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.			
PO5	2.01	2.19	Attainment level improved against target level and hence the same method will be continued for the next year
Action 1: Students exposure to modern tools like Microsoft Azure, GitHub, AWS Cloud 9, Matlab Apache Net beans etc. will be continued for the next academic year also. Action 2: Students will be motivated to do NPTEL and other MOOC courses, seminars, debate, concept map, quiz for next academic year also. Action 3: More motivation by mentors will be done for students to register for Hackathon, Capture the Flag, Paper Presentation, Technical Quiz, Webinars/Seminars conducted by other colleges regarding modern tool usage.			

PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice

PO6	2.01	2.03	Attainment level improved against target level and hence same practice will be followed
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Action 1: Target will be retained and will be observed for next academic year.
Action 2: More exposure of various professional engineering practices followed in the industries for students through industrial visits will be done.
Action 3: Activity based learning will be followed for next academic year also.
Action 4: Association with professional bodies like CSI, ISTE will be followed to create more awareness among the students about professional engineering practice.
Action 5: Continuation of encouragement of students will be done to carry out inter domain projects so that they would realize the importance of a project involving society, safety, health, and the legalities

PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO7	2.01	1.90	Attainment level decreased against target level
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Action 1: Continuation of encouraging students to indulge in mini projects/projects where societal and environmental issues can be addressed will be implemented.
Action 2: Students are strictly instructed to switch off all electrical Equipment /Resources when not in use for all the laboratories.
Action 3: On environmental context Rain water harvesting, Sewage treatment plants, Waste management procedures are employed at our college.

PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO8	2.01	2.06	Attainment level improved against target level and hence same practice will be followed
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Action 1: Professional Ethics will be ensured by giving more importance in scrutinizing the code and project reports of Projects and Mini Projects by doing proper plagiarism check to determine the originality of the project.

PO9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO9	2.01	2.02	Attainment level improved against target level and hence same practice will be followed
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Action 1: Attainment level improved against target level and hence same practice will be followed
Action 2: Encouragement of students to participate in various co-curricular and extra-curricular activities in other colleges/sports activities/cultural activities will be continued.
Action 3: Motivating students by mentors to participate in Inter Collegiate Cultural Fest, Hackathon, Technical Quiz, Paper Presentation will be continued.
Action 4: Continuation of the club activities will be done.

P10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO10	2.01	2.04	Attainment level improved against target level and hence same practice will be followed
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Action 1: Target will be retained and will be observed for next academic year.
Action 2: To enhance the employability skills of the students, the training programs on the topics: mock interview, career development, higher studies motivation talk, GATE coaching classes, GATE videos by faculties ,entrepreneurship development will be continued next academic year also.
Action 3: The Inter/Intra college level project exhibition will be continued for next academic year also.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO11	2.01	2.25	Attainment level improved against target level and hence same practice will be followed
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Action 1: Retention of target will be observed for next academic year.
Action 2: Arrangement will be made like this year for students to visit industries on multidisciplinary environments in the next academic year to improve knowledge of the engineering principles.
Action 4: Like this year students will be encouraged to write case studies as assignments to know about the recent technologies
Action 5: Students will be encouraged to prepare project proposals with the guidance of faculty for government funding agencies like this year.

PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PO12	2.01	2.00	Attainment level decreased against target level
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Action 1: Target attained will be retained
Action 2: Mini Projects in curriculum will be continued which helps the students to perform better in placement interviews and higher studies
Action 3: Like this year students will be motivated to register for GRE/TOEFL/GATE and other competitive examinations.
Action 4: Conduction of GATE coaching classes and GATE coaching videos by faculties will be continued in next academic year also.
Action 5: Students will be encouraged to publish papers in various national and International journals/conferences in next academic year also.
Action 6: Students will be motivated by mentors to register and get certificates in various online courses on new technologies which would help them in continuous learning in next academic year also.

PSO1: To analyze, design and develop computing solutions by applying foundational concepts of Computer Science and Engineering.			
PSO1	2.01	2.03	Attainment level improved against target level and hence same practice will be followed
Action 1: Target attained will be retained Action 2: More demonstrations, case studies, tutorial sessions and quizzes will be adapted into the teaching learning process. Action 3: More ethics related case studies will be given for the knowledge of ethical standards. Action 4: Students will be motivated to do Skill rack, Hackathon, NPTEL courses etc to know efficient solutions for complex technical solutions in next academic year like this year.			
PSO2: To apply software engineering principles and practices for developing quality software for scientific and business applications.			
PSO2	2.01	2.03	Attainment level improved against target level and hence same practice will be followed.
Action 1: Target attained will be retained Action 2: More demonstrations, case studies, tutorial sessions and quizzes will be adapted into the teaching learning process Action 3: Students will be motivated to do Skill rack NPTEL courses etc to know efficient solutions for complex technical solutions in next academic year like this year., Action 4: More guest lectures will be given to students i the industry related technical domains.			
PSO3: To adapt to emerging Information and Communication Technologies (ICT) to innovate ideas and solutions to existing/novel problems.			
PSO3	2.01	1.77	Attainment level decreased against target level
Action 1: Students will be encouraged to earn certifications (e.g., AWS Academy, Google Cloud, Microsoft Azure). Action 2 : Hackathons, Ideathons, and coding challenges to be organized to encourage problem-solving Action 3: Industry experts will be invited for guest lectures and seminars on recent ICT trends and applications. Action 4: More MoUs will be signed with IT industries for internships and collaborative projects.			

Table 7.1.1 POs & PSOs Attainment Levels and Actions